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Department of Mathematics
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The Wall finiteness obstruction

Author: Kamen V. Pavlov (CID: 02347085)

Supervisor: Dr. John Nicholson

The work contained in this thesis is my own work unless otherwise stated.

Abstract

A fundamental problem in algebraic topology is to find conditions to impose on a CW complex to determine precisely when it is homotopy equivalent to one with finitely many cells. The Wall finiteness obstruction is a topological invariant whose vanishing is a necessary and sufficient condition for the existence of such a homotopy equivalence. In this work, we work through the definition of the finiteness obstruction, and we give a self-contained account of the available proof of the major result on its vanishing. We briefly explore the more general problem arising from the finiteness obstruction, the D2 problem, which is a major open problem in topology.

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Introduction

CW complexes are topological spaces built up from unit disks inductively on dimension. This means they are essentially combinatorial objects, and as such are of particular interest to algebraic topology (e.g. as “propagandized for” by Milnor [Mil59]). It is then natural to ask when such a space can be built up from only finitely many unit disks, up to homotopy. More formally:

- When is a CW complex homotopy equivalent to a finite CW complex, i.e. one with finitely many cells?

We say that such a CW complex is *homotopy finite*.

This problem was investigated by Wall [Wal65], who found that there exists an obstruction whose vanishing precisely determines the finiteness of CW complexes up to homotopy. If the space is X , there is an associated abelian group $K_0(\mathbb{Z}[\pi_1(X)])$ from algebraic K-theory (defined in Definition 1.62 below), and the obstruction is an element $\omega(X) \in K_0(\mathbb{Z}[\pi_1(X)])$. Wall [Wal65] proved the following theorem:

- A CW complex X is homotopy equivalent to a finite CW complex if and only if it is finitely dominated and $\omega(X) = 0$

where *finitely dominated* is a property of CW complexes evidently necessary for homotopy finiteness (defined in Definition 1.9 below). It is on such complexes that the finiteness obstruction is defined. However, even defining the obstruction is no menial task. The definition is the sole subject of Chapter 2 of this thesis.

Wall’s original paper [Wal65], as well as its follow-up [Wal66], are excellent mathematical works, containing many ideas and proving multiple results in a far more general setting than just the theorem above requires. They also use nontrivial results implicitly and assume a great many prerequisites (perhaps evidenced by the volume of Chapter 1 of this thesis). As such, these papers are difficult for, e.g., someone at the postgraduate level to read comfortably. Multiple accounts of the finiteness obstruction have been written over the year since the original papers; however these are often no friendlier to the unfamiliar reader, with a similar level of detail and expected background, and they tend to refer back to Wall’s work for the main results [FR01, Theorem 3.1][Ros94, pp. 51–52][Mis95, Theorem 5.2][Ran85, p. 125][Ped17][Nan19]. The book of Varadarajan [Var89] is an entire book solely on the subject of the finiteness obstruction; however, the author has not obtained access to this text and cannot attest to the nature of its contents.

In light of this apparent lack of postgraduate-level accounts of the topic in the literature, the present work aims to serve mainly as a clean, concise, and self-contained treatment of the Wall finiteness obstruction and the proof of the theorem above, starting from as elementary a basis as possible. In order to offer a different point of view to the

geometric account found in e.g. Pedersen [Ped17] and Wall’s original paper [Wal65], this thesis is of a more algebraic flavor, aiming to present a parallel with the algebraic theory of chain complexes. In fact this is what Wall’s second paper [Wal66] is concerned with, and so this work incorporates results and proofs from both papers [Wal65, Wal66]. Also incorporated are results and proofs in the area of homological algebra from the books of Rosenberg [Ros94] and Johnson [Joh12] and the paper of Ranicki [Ran85]. This is done in the hope of providing a simpler and more efficient treatment, as well as offering insight into both the algebraic core of the topic and the more general major open problem, the D2 problem. More discussion on the reasoning behind this choice is included in Section 3.1.

Chapter 1 establishes preliminaries. Chapter 2 defines the Wall finiteness obstruction, which is then utilized in Chapter 3 to prove the main theorem, as stated above. We then conclude with a mention of the D2 problem in Chapter 4.

Assumed knowledge is material covered in introductory postgraduate courses on algebraic topology and homological algebra, corresponding roughly to the first two chapters of Hatcher’s book [Hat02] and the first three chapters of Weibel’s book [Wei94]. Important points will be recalled.

Chapter 1

Preliminaries

In this chapter we will establish the necessary preliminaries. There is a lot of material to get through, so all the proofs in this chapter will be omitted for the sake of brevity. We cover the definitions and very basic theory of CW complexes in Section 1.1. We define the higher homotopy groups of a space in Section 1.2 before going back to CW complexes and cellular homology, including the action of π_1 on invariants of the universal cover, in Section 1.3. Section 1.4 runs through all the necessary homological algebra, and Section 1.5 defines some relevant basic concepts from algebraic K-theory. The main reference for Sections 1.1 to 1.3 is the book by Hatcher [Hat02], while Sections 1.4 and 1.5 draw mainly from Rosenberg's book [Ros94].

1.1 CW complexes

This section incorporates material from various parts of Hatcher's book [Hat02].

Before we can define what a CW complex is, we need to know how to “glue” unit disks together. This will be an example of a general topological construction of attaching one space to another via a continuous map.

Definition 1.1. Let X be a topological space with $A \subseteq X$ a subspace. Suppose that $f : A \rightarrow Y$ is a continuous map to a topological space Y . Then *attaching X to Y by the map f* is the space

$$X \cup_f Y = (X \sqcup Y) / \sim$$

with the quotient topology, where $a \sim f(a)$ for all $a \in A$. f is called an *attaching map*.

Now a CW complex is a space built from attaching unit disks in ascending dimensions in this way, where all disks of the same dimension are attached simultaneously. Formally:

Definition 1.2. A *CW complex* is a topological space X built up inductively in the following way:

- X^0 is an arbitrary set of points N_0 with the discrete topology.
- Construct X^k from X^{k-1} by

$$X^k = \left(\bigsqcup_{i \in N_k} D_i^k \right) \cup_{f_k} X^{k-1}$$

where N_k is an arbitrary set, the D_i^k are k -dimensional disks indexed by N_k , and $f_k = \bigcup_{i \in N_k} f_{k,i}$ is the union of maps from the boundaries of the disks $f_{k,i} : \partial D_i^k \cong S^{k-1} \rightarrow X^{k-1}$.

- X is the space produced at the end of this process, after potentially infinitely many steps.

X^k is called the k -skeleton of X . The D_i^k are called the k -cells of X .

X is *finite-dimensional* if $X = X^n$ for some n , and the smallest such n is the *dimension* of X .

X is said to be *of finite type* if N_k is a finite set for all k .

X is *finite* if it is finite-dimensional and of finite type.

So a CW complex is finite if it consists of finitely many cells, and it can fail to be finite in two ways: there might be infinitely many cells in any one dimensions, or we might have cells in infinitely many dimensions.

Examples of CW complexes include many familiar spaces, such as spheres, tori, and projective spaces. The n -sphere S^n has a CW complex structure with a single 0-cell and a single n -cell whose entire boundary is glued to the 0-cell. Description of a space as a CW complex may not be unique. For more detailed examples, see e.g. Chapter 0 of Hatcher's book [Hat02].

An important example that we will need later is a CW complex associated to a presentation of a group G in the following way:

Definition 1.3. The *presentation complex* for a given group G with presentation $\langle S | R \rangle$ is the 2-dimensional CW complex constructed in the following way:

- Begin with a single 0-cell.
- Attach 1-cells in correspondence with the generators S of G . These form loops α_i indexed by S .
- Now each relation $r = g_1 g_2 \cdots g_k \in R$ specifies a loop $\beta_r = \alpha_{g_1} \cdot \alpha_{g_2} \cdot \alpha_{g_3} \cdots \alpha_{g_k}$ where $\cdot$ denotes composition of loops; since a loop in X can be thought of as a basepoint-preserving map $S^1 \rightarrow X$, we can use the β_r indexed by R as attaching maps for 2-cells.

This complex can be shown to have fundamental group G [Hat02, Corollary 1.28]. In particular, if G is a finitely presented group, there exists a finite complex with fundamental group G . Conversely, we have:

Proposition 1.4 ([Hat02, Proposition 1.26]). For X a finite CW complex, $\pi_1(X)$ is a finitely presented group.

Definition 1.5. A *CW subcomplex* of a CW complex X is a closed subspace $A \subseteq X$ that is a union of cells of X . In this case we sometimes write (X, A) and say that (X, A) is a *CW pair*.

As a main example, each skeleton of X is a subcomplex.

Next we have nice maps between CW complexes:

Definition 1.6. A continuous map $f : X \rightarrow Y$ between CW complexes is *cellular* if $f(X^n) \subseteq Y^n$ for all n .

That is, cellular maps take cells to the same or lower dimension. It is an important theorem that all maps are cellular up to homotopy:

Theorem 1.7 (Cellular Approximation Theorem [Hat02, Theorem 4.8]). Every continuous map between CW complexes is homotopic to a cellular map.

Recall that two spaces X and Y are homotopy equivalent if there exist maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. Our main interest is in the following property of CW complexes:

Definition 1.8. A CW complex X is *homotopy finite* if it is homotopy equivalent to a finite CW complex.

We would like to find conditions to impose on an arbitrary CW complex X to determine precisely when X is homotopy finite. As an easy example of a non-finite CW complex which is homotopy finite, take $\mathbb{R}$. This has the CW complex structure of $\mathbb{Z}$ as 0-cells and the intervals $[n, n+1]$ as 1-cells. However, we know $\mathbb{R}$ to be contractible, i.e. it is homotopy equivalent to the CW complex with just one 0-cell, hence is homotopy finite.

Conversely, we can easily obtain examples of CW complexes which are not homotopy finite: Let G be a group which is not finitely presented, and let X be the presentation complex for G with any presentation. Then Proposition 1.4 tells us that X cannot be homotopy finite.

Now suppose that X is a homotopy finite CW complex, i.e. we have maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ with Y finite such that $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. If we only require one of these homotopies to exist, we get a weaker condition which is evidently necessary for homotopy finiteness.

Definition 1.9. A CW complex X is *finitely dominated* if there exists a finite CW complex K and maps $g : X \rightarrow K$ and $f : K \rightarrow X$ satisfying $f \circ g \simeq \text{id}_X$.

We say that K *dominates* X , or that K is a *(finite) dominating complex* for X . The map f might be called a *homotopy retraction* or a *domination*, and g might be called a *homotopy right-inverse* to f .

In many cases, the dominating complex is a subcomplex. For example, if we have a map $h : X \rightarrow X$ homotopic to the identity and whose image $h(X)$ has compact closure, then a finite subcomplex containing $h(X)$ dominates X [FR01, Proposition 1.3].

It turns out that finitely dominated spaces are convenient for our purposes, in the sense that just one extra relatively simple condition is enough to precisely determine homotopy finiteness: viz., the vanishing of the Wall finiteness obstruction, defined in Chapter 2. In addition, many spaces are known to be finitely dominated, for example all compact topological manifolds (as remarked by Ferry and Ranicki [FR01, Example 1.2]).

Determining homotopy finiteness for disconnected CW complexes can be done separately for each component, so in the sequel we will assume all CW complexes to be connected. This gives us access to universal covers:

Proposition 1.10 ([Hat02, Appendix]). A connected CW complex X is path-connected, locally path-connected, and semi-locally simply-connected; hence X admits a universal cover $\tilde{X}$.

Remark 1.11. We can now easily see, using the lifting criterion [Hat02, Proposition 1.33], that a map $f : X \rightarrow Y$ of CW complexes lifts to a map $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$.

1.2 Higher homotopy groups

This section mostly follows Section 4.1 from Hatcher [Hat02].

Recall that the fundamental group of a space X at the basepoint x_0 is defined as the group of homotopy classes of maps $(I, \partial I) \rightarrow (X, x_0)$ under composition, where I is the unit interval. Alternatively, this can be viewed as the group of homotopy classes of maps $(S^1, s_0) \rightarrow (X, x_0)$, since the circle S^1 is homeomorphic to $I/\partial I$, and we can choose $\partial I/\partial I$ as the basepoint s_0 .

This viewpoint generalizes easily to higher dimensions, so that:

Definition 1.12. For a space X with basepoint x_0 , denote by $\pi_n(X, x_0)$ the set of homotopy classes of maps from the n -sphere $(S^n, s_0) \rightarrow (X, x_0)$.

Proposition 1.13 ([Hat02, pp. 340–341]). $\pi_n(X, x_0)$ forms a group under addition defined by $f + g = (f \vee g) \circ c$, where $c : S^n \rightarrow S^n \vee S^n$ is the map collapsing the equator S^{n-1} to a point, with identity element the homotopy class of the constant map. This group is called the n -th homotopy group of X at x_0 .

Remark 1.14. The basepoint s_0 needs to be chosen to lie on the collapsing equator S^{n-1} .

The higher homotopy groups share many of the nice properties of π_1 . For one, different choices of basepoint result in isomorphic groups as long as X is path-connected, so we can suppress the basepoint in the notation henceforth, writing $\pi_n(X)$. Also, π_n is a functor, with induced homomorphisms $f_* : \pi_n(X) \rightarrow \pi_n(Y)$ defined for $f : X \rightarrow Y$ by composition the same way as with π_1 .

Unlike the fundamental group however, the higher homotopy groups are all abelian, as (informally) the higher dimensions give us space to “switch the places” of f and g via homotopy. Another difference is that the higher homotopy groups are a lot more difficult to compute in general. See Hatcher’s Section 4.1 [Hat02] for detailed treatment of the higher homotopy groups.

An easy computation that we can make is via the cellular approximation theorem, which immediately gives that any map $S^n \rightarrow S^k$ for $n < k$ is nullhomotopic, hence $\pi_n(S^k) = 0$.

Definition 1.15. A map $f : X \rightarrow Y$ of topological spaces is a *weak homotopy equivalence* if it induces isomorphisms on all homotopy groups, i.e. if $f_* : \pi_n(X) \rightarrow \pi_n(Y)$ is a group isomorphism for all $n \geq 1$.

The following theorem shows one of the reasons why CW complexes are so important in algebraic topology, and will be one of our main tools in building a finite CW complex homotopy equivalent to a given finitely dominated one.

Theorem 1.16 (Whitehead's Theorem [Hat02, Theorem 4.5]). A weak homotopy equivalence between CW complexes is a homotopy equivalence.

We also have the following important result:

Theorem 1.17 ([Hat02, Proposition 4.1]). Covering spaces $p : \tilde{X} \rightarrow X$ induce isomorphisms $\tilde{p} : \pi_n(\tilde{X}) \rightarrow \pi_n(X)$ for all $n \geq 2$.

Next we define the relative version of the homotopy groups:

Definition 1.18. For a pair (X, A) with basepoint x_0 (i.e. $x_0 \in A \subseteq X$), denote by $\pi_n(X, A, x_0)$ the set of homotopy classes of maps $(D^n, S^{n-1}, s_0) \rightarrow (X, A, x_0)$.

Proposition 1.19 ([Hat02, p. 343]). $\pi_n(X, A, x_0)$ forms a group under addition defined by $f + g = (f \vee g) \circ c$, where $c : D^n \rightarrow D^n \vee D^n$ is the map collapsing the equator D^{n-1} to a point, with identity element the homotopy class of the constant map. This group is called the *n-th relative homotopy group* of (X, A) at x_0 .

Remark 1.20. For $n = 1$ addition is not well-defined, so $\pi_1(X, A, x_0)$ is just a set.

Again we suppress the basepoint whenever A and X are path-connected.

$\pi_n(X, A)$ is also a functor: $f : (X, A) \rightarrow (Y, B)$ induces $f_* : \pi_n(X, A) \rightarrow \pi_n(Y, B)$. Similarly to π_n for $n \geq 2$, $\pi_n(X, A)$ is abelian for $n \geq 3$. (For details see Section 4.1 in Hatcher [Hat02].)

Recall that there are homology long exact sequences associated to pairs of spaces (X, A) as well as triples of spaces (X, A, B) (i.e. $B \subseteq A \subseteq X$), where we are using the singular construction on homology as given in Chapter 2 of Hatcher [Hat02]. There are similar long exact sequences using relative homotopy groups:

Theorem 1.21 ([Hat02, Theorem 4.3]). For a triple (X, A, B) , the relative homotopy groups fit into the long exact sequence

$$\cdots \xrightarrow{\partial} \pi_n(A, B) \xrightarrow{\iota_*} \pi_n(X, B) \xrightarrow{j_*} \pi_n(X, A) \xrightarrow{\partial} \pi_{n-1}(A, B) \xrightarrow{\iota_*} \cdots \xrightarrow{j_*} \pi_1(X, A),$$

where $\iota : (A, B) \rightarrow (X, B)$ and $j : (X, B) \rightarrow (X, A)$ are inclusions, and ∂ is the composition $\pi_n(X, A) \rightarrow \pi_{n-1}(A) \rightarrow \pi_{n-1}(A, B)$ where the second map is induced by the inclusion and the first comes from restricting maps $(D^n, S^{n-1}) \rightarrow (X, A)$ to S^{n-1} .

Remark 1.22. At the end of the sequence the terms are sets and not groups, but exactness still makes sense by taking kernels to mean those elements which map to the homotopy class of the constant map.

Applying this theorem to the case when B is just the basepoint yields:

Corollary 1.23. For a pair (X, A) , we have the homotopy exact sequence

$$\cdots \rightarrow \pi_n(A) \rightarrow \pi_n(X) \rightarrow \pi_n(X, A) \rightarrow \pi_{n-1}(A) \rightarrow \cdots \rightarrow \pi_1(X, A).$$

Generalizing the simply-connected property of spaces, we have:

Definition 1.24. A space X is *n-connected* if $\pi_i(X) = 0$ for all $i \leq n$.

A pair (X, A) is *n-connected* if $\pi_i(X, A) = 0$ for all $i \leq n$.

Remark 1.25. Using the long exact sequence, we can check that (X, A) being n -connected is equivalent to the inclusion $A \rightarrow X$ inducing isomorphisms on π_i for $i < n$ and a surjection on π_n .

The cellular approximation theorem gives that, for X a CW complex, any map $(D^i, S^{i-1}) \rightarrow (X, X^n)$ for $i \leq n$ is nullhomotopic, hence:

Proposition 1.26 ([Hat02, Corollary 4.12]). For X a CW complex, the pair (X, X^n) is n -connected.

Recall that the mapping cylinder of a map $f : X \rightarrow Y$ is a space M_f containing both X and Y as subspaces and which is homotopy equivalent to Y . Hence, as long as we are only considering spaces up to homotopy equivalence, we can regard any map as an inclusion by replacing Y with the mapping cylinder. In particular, this allows us to define homotopy groups associated to maps, as Wall does [Wal65, p. 56]:

Definition 1.27. The n -th homotopy group of a map $f : X \rightarrow Y$ is $\pi_n(f) := \pi_n(M_f, X)$. A map f is n -connected if $\pi_i(f) = 0$ for all $i \leq n$.

The n -th homology group of a map $f : X \rightarrow Y$ is $H_n(f) := H_n(M_f, X)$.

Now since M_f is homotopy equivalent to Y they have the same homotopy and homology groups and the inclusion $X \rightarrow M_f$ induces the same maps on homotopy and homology as f , so:

Definition 1.28. For a map $f : X \rightarrow Y$, we have the *homotopy and homology long exact sequences of f* :

$$\begin{aligned} \cdots \rightarrow \pi_n(X) \xrightarrow{f_*} \pi_n(Y) \rightarrow \pi_n(f) \rightarrow \pi_{n-1}(X) \xrightarrow{f_*} \cdots \rightarrow \pi_1(f), \\ \cdots \rightarrow H_n(X) \xrightarrow{f_*} H_n(Y) \rightarrow H_n(f) \rightarrow H_{n-1}(X) \xrightarrow{f_*} \cdots \rightarrow H_0(f). \end{aligned}$$

Remark 1.29. The homotopy exact sequence shows that f being n -connected is equivalent to f inducing isomorphisms on π_i for $i < n$ and a surjection on π_n .

As a consequence of Theorem 1.17 and the homotopy exact sequence of a map, we have that:

Proposition 1.30. If $f : X \rightarrow Y$ is a map between CW complexes, $\pi_n(f) \cong \pi_n(\tilde{f})$ for $n \geq 3$. The same is true for $n = 1, 2$ if f induces an isomorphism on fundamental groups.

We finish this section by stating a version of one of the most important theorems in algebraic topology, relating the homotopy and homology groups of a space. This will be used a lot in the sequel, at times implicitly.

Theorem 1.31 (The Hurewicz Theorem [Hat02, Theorem 4.32]). (i) For an n -connected space X with $n \geq 1$, we have that $H_i(X) = 0$ for $1 \leq i \leq n$ and that $\pi_{n+1}(X) \cong H_{n+1}(X)$.

(ii) For an n -connected pair (X, A) with $n \geq 1$ and A simply-connected and nonempty, we have that $H_i(X, A) = 0$ for $i \leq n$ and that $\pi_{n+1}(X, A) \cong H_{n+1}(X, A)$.

(iii) For an n -connected map $f : X \rightarrow Y$ with $n \geq 1$ and X simply-connected, we have that $H_i(f) = 0$ for $i \leq n$ and that $\pi_{n+1}(f) \cong H_{n+1}(f)$.

So for a simply-connected space X we have the first nonzero homotopy and homology groups occurring in the same dimension, and they are isomorphic. The same is true for pairs (X, A) with $\pi_1(X, A) = 0$ and A simply-connected, and for maps $f : X \rightarrow Y$ with $\pi_1(f) = 0$ and X simply-connected.

Combining the Hurewicz theorem with Whitehead's theorem, we can see:

Corollary 1.32 ([Hat02, Corollary 4.33]). A map between simply-connected CW complexes which induces isomorphisms on all homology groups is a homotopy equivalence.

1.3 Cellular homology and the action of π_1

Main references for this section are Section 2.2 in Hatcher [Hat02] and Section 5.2.1 in the lecture notes of Davis and Kirk [DK01].

Another big reason CW complexes are important in algebraic topology is that there exists a tool, called cellular homology, which among other things makes the computation of homology groups really efficient. To define it, we first need a small theorem.

Theorem 1.33 ([Hat02, Lemma 2.34]). If X is a CW complex, then for all n , we have that $H_n(X^n, X^{n-1})$ is a free abelian group with basis the n -cells of X .

Then we can fit these free abelian groups into a sequence as follows:

Definition 1.34. Let X be a CW complex. Then denote by $C_*(X)$ the sequence

$$\cdots \xrightarrow{d_{n+2}} H_{n+1}(X^{n+1}, X^n) \xrightarrow{d_{n+1}} H_n(X^n, X^{n-1}) \xrightarrow{d_n} H_{n-1}(X^{n-1}, X^{n-2}) \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} H_0(X^0),$$

where d_i is the composition $H_i(X^i, X^{i-1}) \rightarrow H_{i-1}(X^{i-1}) \rightarrow H_{n-1}(X^{i-1}, X^{i-2})$ with the maps coming from the homology exact sequences of the pairs (X^i, X^{i-1}) and (X^{i-1}, X^{i-2}) respectively. Denote also $C_i(X) = H_i(X^i, X^{i-1})$.

The elements of $H_i(X^i, X^{i-1})$ are called the *cellular i -chains* of X , and the maps d_i are the *cellular boundary maps*.

Remark 1.35. The notation $C_*(X)$ and $C_i(X)$ has potential to cause confusion, as these are used for the singular complex and singular i -chains of a space. However, we won't need to refer to the singular complex and chains much; when we do, we will instead use $S_*(X)$ and $S_i(X)$.

The remarkable fact is then that the following theorem holds.

Theorem 1.36 ([Hat02, Theorem 2.35]). $C_*(X)$ is a chain complex and we have that

$$H_i(C_*(X)) \cong H_i(X)$$

where $H_i(X)$ are the ordinary singular homology groups of X .

In light of this theorem, call $C_*(X)$ the *cellular chain complex* of X , the elements of $C_i(X)$ the *cellular i -chains* of X , and the maps d_i of $C_*(X)$ the *cellular boundary maps*. The homology groups of X can be computed with $C_*(X)$ in place of the usual $S_*(X)$: from now on, $H_i(X)$ will be used for both, with the isomorphism implicit.

Now, as the $C_i(X)$ are free abelian groups with bases the cells of X , computing homology becomes a lot less unwieldy. For example, since subgroups of finitely generated abelian groups are finitely generated, we immediately get that:

Proposition 1.37. For X a finite CW complex, $H_i(X)$ is a finitely presented abelian group for all i .

The cellular boundary maps also have a nice description in terms of degree for maps $S^n \rightarrow S^n$ [Hat02, Section 2.2, Cellular Boundary Formula], but we won't need this here.

Now recall from Proposition 1.10 that a CW complex X admits a universal cover $\tilde{X}$, and that $\pi_1(X)$ acts freely on $\tilde{X}$ by deck transformations. This gives us:

Definition 1.38. Let X be a CW complex. Then $\pi_1(X, x_0)$ acts on $\pi_n(\tilde{X}, \tilde{x}_0)$ for $n \geq 2$ as follows: Given $g \in \pi_1(X, x_0)$, let $g' : \tilde{X} \rightarrow \tilde{X}$ be the associated deck transformation, sending $\tilde{x}_0$ to $\tilde{x}_1$. Then

$$g \cdot \alpha := \beta(g'_*(\alpha)),$$

where $g'_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(\tilde{X}, \tilde{x}_1)$ is induced by g' and $\beta : \pi_1(\tilde{X}, \tilde{x}_1) \rightarrow \pi_1(\tilde{X}, \tilde{x}_0)$ is a change of basepoint transformation.

We can easily check this action to be well defined and linear with respect to addition in $\pi_n(\tilde{X})$. Hence it defines the following:

Definition 1.39. For a CW complex X , the n -th homotopy module of the universal cover of X is $\pi_n(\tilde{X})$ with the $\mathbb{Z}[\pi_1(X)]$ -module structure given by the action of $\pi_1(X)$ on $\pi_n(\tilde{X})$.

The n -th homotopy module of X is $\pi_n(X)$ with the $\mathbb{Z}[\pi_1(X)]$ -module structure induced by the isomorphism $\pi_n(X) \cong \pi_n(\tilde{X})$ from Theorem 1.17.

Remark 1.40. The definition only works for $n \geq 2$, as we need $\pi_n(\tilde{X})$ to be abelian in order to make it into a $\mathbb{Z}[\pi_1(X)]$ -module.

Similarly, if we have a CW pair $(\tilde{X}, \tilde{A})$, we have a linear action of $\pi_1(A)$ on $\pi_n(X, A)$; for details of this, see Davis and Kirk [DK01, Theorem 6.63]. We will only be interested in cases when the inclusion induces an isomorphism on fundamental groups, hence we give a simplified definition:

Definition 1.41. Let (X, A) be a CW pair and let the inclusion $A \rightarrow X$ induce an isomorphism $\pi_1(A) \cong \pi_1(X)$.

The n -th relative homotopy module of the universal covers of (X, A) is $\pi_n(\tilde{X}, \tilde{A})$ with the $\mathbb{Z}[\pi_1(X)]$ -module structure given by the action of $\pi_1(X)$ on $\pi_n(\tilde{X}, \tilde{A})$.

The n -th relative homotopy module of (X, A) is $\pi_n(X, A)$ with the $\mathbb{Z}[\pi_1(X)]$ -module induced by the isomorphism $\pi_n(X, A) \cong \pi_n(\tilde{X}, \tilde{A})$ from Proposition 1.30.

Remark 1.42. The definition only works for $n \geq 3$, as we need $\pi_n(\tilde{X}, \tilde{A})$ to be abelian in order to make it into a $\mathbb{Z}[\pi_1(X)]$ -module.

We can also similarly define the homotopy modules $\pi_n(f)$ and $\pi_n(\tilde{f})$ of a map $f : X \rightarrow Y$ using the mapping cylinder of f as before, and the definition will make sense as long as f induces isomorphisms on fundamental groups.

Theorem 1.43 ([DK01, Theorem 6.6.3]). (i) Induced maps of homotopy modules are homomorphisms of $\mathbb{Z}[\pi_1(X)]$ -modules.

(ii) For a CW triple (X, A, B) with all inclusions inducing isomorphisms on fundamental groups, all maps in the homotopy exact sequence in Theorem 1.21 are homomorphisms of $\mathbb{Z}[\pi_1(X)]$ -modules.

(iii) The same holds for the homotopy exact sequence of a pair (X, A) with the inclusion inducing an isomorphism on fundamental groups.

(iv) The same holds for the homotopy exact sequence of a map $f : X \rightarrow Y$ with f inducing an isomorphism on fundamental groups.

Now let's see that we have a similar situation with homology. The free action of $\pi_1(X)$ on $\tilde{X}$ induces an action on the singular simplices of $\tilde{X}$, which is then extended linearly to $S_n(\tilde{X})$. Then:

Definition 1.44. The n -th homology module of the universal cover of a CW complex X is $H_n(S_*(\tilde{X}))$ where $S_*(\tilde{X})$ is considered as a complex of $\mathbb{Z}[\pi_1(X)]$ -modules.

$\mathbb{Z}[\pi_1(X)]$ -module structures on $H_n(X, A)$ and $H_n(f)$ are defined intuitively, as long as the inclusion $A \rightarrow X$ and f induce isomorphisms on fundamental groups.

Remark 1.45. There are no homology modules associated to X itself.

Many of the maps seen in Section 1.2 (including induced homomorphisms $\tilde{f}_* : \tilde{X} \rightarrow \tilde{Y}$ and maps in homology exact sequences of universal covers) can be seen to be homomorphisms of $\mathbb{Z}[\pi_1(X)]$ -modules by various homological algebra results.

We also have the module version of the Hurewicz theorem, which will also be used often and at times without mention:

Proposition 1.46 (The Hurewicz Theorem for modules). (i) For a CW complex X with n -connected $\tilde{X}$, we have that $H_i(\tilde{X}) = 0$ for $1 \leq i \leq n$ and $\pi_{n+1}(X) \cong \pi_{n+1}(\tilde{X}) \cong H_{n+1}(\tilde{X})$ as $\mathbb{Z}[\pi_1(X)]$ -modules.

(ii) For a CW pair (X, A) with the inclusion inducing an isomorphism on fundamental groups and $(\tilde{X}, \tilde{A})$ n -connected, we have that $H_i(\tilde{X}, \tilde{A}) = 0$ for $i \leq n$ and that $\pi_{n+1}(X, A) \cong \pi_{n+1}(\tilde{X}, \tilde{A}) \cong H_{n+1}(X, A)$ as $\mathbb{Z}[\pi_1(X)]$ -modules.

(iii) For a map $f : X \rightarrow Y$ of CW complexes inducing an isomorphism on fundamental groups and $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ n -connected, we have that $H_i(\tilde{f}) = 0$ for $i \leq n$ and that $\pi_{n+1}(f) \cong \pi_{n+1}(\tilde{f}) \cong H_{n+1}(f)$ as $\mathbb{Z}[\pi_1(X)]$ -modules.

Remark 1.47. Precise references for this result—or any more general result—seem hard to come by. The absolute version appears (without proof) as (19.5) in a paper of Eilenberg and MacLane [EM49], which in turn refers to the original paper of Hurewicz [Hur95, pp. 364–373]. However, the result seems ubiquitous and is used freely by Wall [Wal65, Wal66], as well as others [Joh12, pp. x, 162]. Indeed, all we need to do is prove that the isomorphism given by the Hurewicz theorem is a homomorphism of $\mathbb{Z}[\pi_1(X)]$ -modules. This seems trivial from the definitions; however, as we have not given the definition of the Hurewicz map, we do not provide details.

In the sequel, whenever X is a CW complex and we write $\pi_n(X)$, $\pi_n(\tilde{X})$, $H_n(\tilde{X})$, etc. (including versions of these for pairs and maps), we will have the module structure in mind, unless explicitly stated that we mean the underlying group. Similarly, for $C_*(\tilde{X})$ we will have in mind the structure of a chain complex of $\mathbb{Z}[\pi_1(X)]$ -modules; maps and isomorphisms will be taken to mean homomorphisms and isomorphisms of $\mathbb{Z}[\pi_1(X)]$ -modules unless otherwise stated; and language like “finitely generated” will mean as a $\mathbb{Z}[\pi_1(X)]$ -module instead of as a group unless noted.

Now let's refer back to the groups of cellular chains of a CW complex.

Proposition 1.48 (See Section 5.2.1 in Davis and Kirk [DK01]). $\tilde{X}$ inherits a CW structure from the CW structure of X : the n -cells of $\tilde{X}$ are the path-components of the preimage of each n -cell of X under the covering space map $\tilde{X} \rightarrow X$.

Hence we know by Theorem 1.33 that $C_n(\tilde{X}) = H_n(\tilde{X}^n, \tilde{X}^{n-1})$ is a free abelian group with basis the n -cells of $\tilde{X}$. In general, this won't be finitely generated even for a finite CW complex X , as that would require the group $\pi_1(X)$ to be finite as well. However, $H_n(\tilde{X}^n, \tilde{X}^{n-1})$ is actually a $\mathbb{Z}[\pi_1(X)]$ -module by our discussion above, and it's not hard to see, using the fact the action of $\pi_1(X)$ on $\tilde{X}$ is free, that we have:

Proposition 1.49. If X is a CW complex, then for all n , we have that $H_n(\tilde{X}^n, \tilde{X}^{n-1})$ is a free $\mathbb{Z}[\pi_1(X)]$ -module with basis the n -cells of X .

We have now an improved version of Theorem 1.36:

Proposition 1.50 (See Section 5.2.1 in Davis and Kirk [DK01]). $C_*(\tilde{X})$ is a chain complex of free $\mathbb{Z}[\pi_1(X)]$ -modules and we have that

$$H_i(C_*(\tilde{X})) \cong H_i(X)$$

as $\mathbb{Z}[\pi_1(X)]$ -modules.

Remark 1.51. Note that, unlike in Proposition 1.37, this doesn't give us that $H_i(\tilde{X})$ are finitely presented $\mathbb{Z}[\pi_1(X)]$ -modules for a finite X . This is because, unlike with abelian groups, submodules of finitely generated modules need not be finitely generated, even for free modules. In fact, an example of a finite CW complex X whose $H_i(\tilde{X})$ is not even finitely generated can be easily derived from a paper of Stallings [Sta63] and also from examples in Hatcher [Hat02, Section 4.A]. This will be discussed further in Section 2.2.

1.4 Chain complexes

This section recalls standard homological algebra material, and as such there is no specific reference. However, most of the material is covered concisely (though at a different level of generality) in Section 1.7 of Rosenberg [Ros94].

For this section and the next, R is a fixed ring with unity. Recall that a chain complex C_* is a sequence of R -modules with differential maps

$$\cdots \xrightarrow{d} C_{n+1} \xrightarrow{d} C_n \xrightarrow{d} C_{n-1} \xrightarrow{d} \cdots$$

such that $d^2 = 0$. The chain complexes we encounter will mainly be cellular complexes of CW complexes, so we will assume all chain complexes to start at dimension 0:

$$\cdots \xrightarrow{d} C_n \xrightarrow{d} C_{n-1} \xrightarrow{d} \cdots \xrightarrow{d} C_0.$$

To draw a parallel with the terminology for CW complexes, we will use the following language for chain complexes:

- C_* is *n-dimensional* if $C_i = 0$ for all $i > n$ but $C_n \neq 0$. C_* is *finite-dimensional* if it is n -dimensional for some n .

- C_* is said to be *of finite type* if all C_i are finitely generated.
- C_* is *finite* if it is finite-dimensional and of finite type.
- The n -*skeleton* of C_* is the chain complex obtained by truncating C_* at dimension n , i.e.

$$C_*^n = C_n \xrightarrow{d} \cdots \xrightarrow{d} C_0.$$

In addition, we will say that C_* is free (respectively projective) if C_i is free (respectively projective) for all i . For example, take a finite CW complex X : then $C_*(\tilde{X})$ is a finite free chain complex of $\mathbb{Z}[\pi_1(X)]$ -modules.

Next recall that a chain map $f : C_* \rightarrow C'_*$ is a collection of maps $f : C_n \rightarrow C'_n$ such that $d'f = fd$. Chain maps induce maps on homology $f_* : H_n(C_*) \rightarrow H_n(C'_*)$.

Definition 1.52. If $f, g : C_* \rightarrow C'_*$ are two chain maps, a *chain homotopy* between them $h : f \simeq g$ is a collection of maps $h_n : C_n \rightarrow C'_{n+1}$ such that $hd + d'h = f - g$.

A chain map $f : C_* \rightarrow C'_*$ is a *chain homotopy equivalence* if there exists a chain map $g : C'_* \rightarrow C_*$ such that $fg \simeq \text{id}_{C'_*}$ and $gf \simeq \text{id}_{C_*}$. In this case C_* and C'_* are *chain homotopy equivalent*, and they're said to be *of the same homotopy type*, and we write $C_* \simeq C'_*$. Chain homotopy equivalences induce isomorphisms on homology.

Finally, a chain complex C_* is *chain contractible* if there exists a chain homotopy between the zero map and the identity map on C_* .

With all the above terms, we might sometimes drop the words “chain”, “homotopy”, or both, if the context is clear enough. E.g. “chain equivalence” might mean a chain homotopy equivalence, or “homotopy” might mean a chain homotopy.

Recall from basic homology theory that a map $f : X \rightarrow Y$ of spaces induces a chain map $f_* : S_*(X) \rightarrow S_*(Y)$. Also $f : (X^n, X^{n-1}) \rightarrow (Y^n, Y^{n-1})$ can be taken to be cellular by the cellular approximation theorem, hence f induces maps on cellular chains, and thus f also induces chain maps $f_* : C_*(X) \rightarrow C_*(Y)$ and $\tilde{f}_* : C_*(\tilde{X}) \rightarrow C_*(\tilde{Y})$.

We will be paying special attention to chain complexes with the homotopy type of a finite projective chain complex. We give them a name:

Definition 1.53. A chain complex C_* is *finitely dominated* if it is chain homotopy equivalent to a finite projective chain complex.

Remark 1.54. One might object that it is more sensible to give the name “finitely dominated” to chain complexes obtained by analogy with finitely dominated CW complexes, i.e. if there exist chain maps $f : C_* \rightarrow F_*$ and $g : F_* \rightarrow C_*$ with F_* finite and free such that $gf \simeq \text{id}_{C_*}$. Indeed, the two concepts are equivalent, as shown by Ranicki [Ran85, Proposition 3.1]. However, for our purposes, we refer to finitely dominated complexes in the sense of Definition 1.53 more often, and we won't need the full generality of Ranicki's result, electing instead to show only that $C_*(\tilde{X})$ is finitely dominated (in the sense of Definition 1.53) for a finitely dominated CW complex X in Theorem 2.1 below. This amounts to showing one direction of Ranicki's result in the special case of integral group rings.

Given two chain complexes C_* and C'_* , one can form their direct sum $C_* \oplus C'_*$ as follows:

$$\cdots \xrightarrow{d \oplus d'} C_n \oplus C'_n \xrightarrow{d \oplus d'} C_{n-1} \oplus C'_{n-1} \xrightarrow{d \oplus d'} \cdots \xrightarrow{d \oplus d'} C_0 \oplus C'_0.$$

If C'_* is contractible, then C_* is seen to be chain equivalent to $C_* \oplus C'_*$ via the inclusion map ι , since the homotopy $\text{id}'_{C'_*} \simeq 0$ gives rise to a homotopy between $\text{id}_{C_* \oplus C'_*}$ and the map $C_* \oplus C'_* \xrightarrow{(\text{id}, 0)} C_* \xrightarrow{\iota} C_* \oplus C'_*$. So a way to alter chain complexes while preserving their homotopy type is to form the direct sum with contractible complexes. To this end, the following very simple contractible complexes serve as “building blocks”:

Definition 1.55. An *elementary chain complex* is a chain complex of the form

$$A \xrightarrow{\text{id}} A \rightarrow 0 \rightarrow \cdots \rightarrow 0,$$

where A is an arbitrary R -module. We denote such complexes by $(A)_*^n$, where $n \geq 1$ is the dimension.

Remark 1.56. This definition, along with the name and notation, is not standard. Despite the utility of elementary complexes, they seem to be too ubiquitous to have been given a name or notation in the literature, where they are always used either implicitly or with direct description. The one exception is a paper of Ranicki [Ran85, p. 115], and this is where the definition is lifted from.

Elementary complexes are clearly contractible, via the homotopy consisting of identity maps. Hence by the comments above, for any R -module A and any $n \geq 1$, we have $C_* \simeq C_* \oplus (A)_*^n$.

Next we introduce the following chain complex associated to a chain map:

Definition 1.57. For a chain map $f : C_* \rightarrow C'_*$, the *algebraic mapping cone* of f , denoted M_*^f , is the chain complex

$$\cdots \xrightarrow{d''_{n+1}} C'_n \oplus C_{n-1} \xrightarrow{d''_n} C'_{n-1} \oplus C_{n-2} \xrightarrow{d''_{n-1}} \cdots \xrightarrow{d''_1} C'_0,$$

where $d''_i(c', c) = (d'_i(c') + f(c), -d_{i-1}(c))$.

Remark 1.58. The name comes from the fact that the topological mapping cone C_f of a map of spaces $f : X \rightarrow Y$ has singular complex $S_*(C_f)$ chain homotopy equivalent to the algebraic mapping cone M_*^{f*} of the induced chain map $f_* : S_*(X) \rightarrow S_*(Y)$ [Wei94, pp. 19–20].

The algebraic mapping cone has some nice properties:

Theorem 1.59 ([Ros94, Theorem 1.7.7]). A chain map $f : C_* \rightarrow C'_*$ is a chain homotopy equivalence if and only if M_*^f is chain contractible. If C_* and C'_* are projective, the following are equivalent:

- f is a chain homotopy equivalence;
- M_*^f is chain contractible;
- M_*^f is exact;
- f induces isomorphisms on all homology modules.

Corollary 1.60. If $f : X \rightarrow Y$ is a homotopy equivalence, then $\tilde{f}_* : C_*(\tilde{X}) \rightarrow C_*(\tilde{Y})$ is a chain equivalence.

Theorem 1.61. Suppose $f : X \rightarrow Y$ is a map of CW complexes and $f_* : C_*(X) \rightarrow C_*(Y)$ is the induced chain map. Then

$$H_n(f) \cong H_n(M_*^{f_*})$$

as abelian groups for all n . Further, if f induces an isomorphism on fundamental groups, let $\tilde{f}_* : C_*(\tilde{X}) \rightarrow C_*(\tilde{Y})$ be the induced chain map of $\mathbb{Z}[\pi_1(X)]$ -module complexes. Then

$$H_n(\tilde{f}) \cong H_n(M_*^{\tilde{f}_*})$$

as $\mathbb{Z}[\pi_1(X)]$ -modules for all n .

The proof is easy from the long exact sequences and the five lemma. (See the proof of Theorem 1.59 [Ros94, Theorem 1.7.7] for a description of the long exact sequence associated to M_*^f .)

We finish the section with a short comment on cohomology. We shan't dwell on the subject, since we will only need it at one small but crucial point, but we mention the main points: For any chain complex C_* and any R -module A , we can form the dual cochain complex of abelian groups

$$\mathrm{Hom}_R(C_0, A) \xrightarrow{\partial} \cdots \xrightarrow{\partial} \mathrm{Hom}_R(C_{n-1}, A) \xrightarrow{\partial} \mathrm{Hom}_R(C_n, A) \xrightarrow{\partial} \cdots,$$

where ∂ is precomposition with d . The cohomology groups $H^i(C_*, A)$ are then defined analogously to the homology groups of a chain complex. Chain maps induce homomorphisms on cohomology groups, while chain homotopy equivalences induce isomorphisms. If X is a CW complex, we put $H^i(X; A) := H^i(C_*(X), A)$ for an abelian group A and $H^i(\tilde{X}; B) := H^i(C_*(\tilde{X}), B)$ for a $\mathbb{Z}[\pi_1(X)]$ -module B . In particular, if X is a finitely dominated CW complex and n is the dimension of the dominating complex, $H^i(X; A) = 0$ and $H^i(\tilde{X}; B) = 0$ for all $i > n$ and any choice of A and B . For a full treatment of cohomology, see Chapter 3 in Hatcher [Hat02].

1.5 Algebraic K-theory and Euler class

The main references for this section is Chapter 1 of [Ros94].

Recall that projective R -modules are direct summands of free R -modules. In particular, for every finitely generated projective module P , we can choose a submodule $[P]$ of R^n for some n which is isomorphic to P . Taking these as representatives of the isomorphism classes of finitely generated projective modules, we see that they form a set. Using this, and noting that this set is closed under taking direct sum (up to isomorphism), we can make the following definition of a group associated to a R :

Definition 1.62. The *projective class group* of R , denoted by $K_0(R)$, is the abelian group with generators the isomorphism classes of finitely generated projective R -modules and with relations $[P \oplus Q] = [P] + [Q]$ for all finitely generated projective R -modules P, Q .

The *reduced projective class group* $\tilde{K}_0(R)$ is the cokernel of the group homomorphism $\mathbb{Z} \rightarrow K_0(R)$ given by $n \mapsto [R^n]$.

Remark 1.63. We can easily check that $[P] = 0 \in \tilde{K}_0(R)$ if and only if P is stably free. Hence $\tilde{K}_0(R)$ in some sense measures the failure of finitely generated projective modules to be stably free.

Recall that in a PID all projective modules are free, hence $\widetilde{K}_0$ is trivial. The same is true for local rings [Ros94, Theorem 1.3.11].

Now if we have a finite projective chain complex P_* , each P_n determines an element of $K_0(R)$, and we can make the following definition:

Definition 1.64. The *Euler class* of a finite projective chain complex P_* is the element of $K_0(R)$ defined by

$$\chi(P_*) = \sum_{i=0}^{\infty} (-1)^i [P_i] \in K_0(R),$$

where we write the sum as infinite for convenience, but since P_* is finite-dimensional, only finitely many terms in the sum are nonzero.

The *reduced Euler class* of P_* is the image of $\chi(P_*)$ in $\widetilde{K}_0(R)$, denoted by $\widetilde{\chi}(P_*)$.

We will see later (Theorem 2.3) that the Euler class (and hence the reduced Euler class) is invariant under chain homotopy equivalence, in some sense justifying the definition. This also allows us to define the (reduced) Euler class unambiguously for any finitely dominated chain complex: this is one of the reasons we use Definition 1.53. The reduced Euler class will be the main ingredient in the definition of the Wall finiteness obstruction.

We can see easily that, if F_* is a finite free chain complex, then $\widetilde{\chi}(F_*) = 0$. The converse also holds, and this will be an important step in the proof of our main result Theorem 3.1, so we will prove it in Theorem 3.3 below.

Chapter 2

Defining the obstruction

In this chapter we will define the Wall finiteness obstruction for a finitely dominated CW complex X . This is no simple task however, as there are a number of things to do along the way. We begin with Section 2.1 in which we describe a general construction needed for the later proofs. Then we will prove:

Theorem 2.1 ([Wal66, Theorem 6]). If a CW complex X is finitely dominated, then $C_*(\tilde{X})$ is a finitely dominated chain complex.

Remark 2.2. We remark here again that there is a purely algebraic proof of Theorem 2.1 contained in a paper of Ranicki [Ran85]. In fact Ranicki's proof is more general, as explained in Remark 1.54. We include instead the original, more geometric proof of Wall [Wal65, Wal66], since it is more akin to the rest of the proofs we go through, and it better illuminates the underlying geometry.

We will prove this in two steps: first showing $C_*(\tilde{X})$ has the homotopy type of a free complex of finite type in Section 2.2 before altering this complex to a finitely dominated one in Section 2.3. In Section 2.4 we check that the Euler class is a homotopy invariant:

Theorem 2.3 ([Ros94, Corollary 1.7.11]). If P_* and P'_* are two homotopy equivalent finite projective chain complexes, then $\chi(P_*) = \chi(P'_*)$.

And with this we can unambiguously define the Euler class for any finitely dominated chain complex:

Definition 2.4. The *Euler class* of a finitely dominated chain complex C_* is defined by $\chi(C_*) = \chi(P_*)$, where P_* is any finite projective complex equivalent to C_* . By Theorem 2.3 this does not depend on the choice of P_* . The *reduced Euler class* of C_* is the image of $\chi(C_*)$ in $\tilde{K}_0(R)$, denoted by $\tilde{\chi}(C_*)$.

Finally, combining this with Theorem 2.1, we can define the Wall finiteness obstruction. We use the algebraic reformulation [Wal66, p. 138] as opposed to the earlier definition [Wal65, Theorem F]; this is also the definition used by Mislin [Mis95, p. 1268]:

Definition 2.5. For a finitely dominated CW complex X , the *Wall finiteness obstruction* is defined by

$$\omega(X) = \tilde{\chi}(C_*(\tilde{X})) \in \tilde{K}_0(\mathbb{Z}[\pi_1(X)]).$$

We might at times refer to the Wall finiteness obstruction as *the finiteness obstruction* or simply *the obstruction*.

Throughout this chapter and the next, we mainly present results from Wall's papers [Wal65, Wal66]. We mostly reproduce the proofs of the statements as referenced, but with added details. Wall is concerned with a higher level of generality and several different finiteness conditions of CW complexes; we focus just on the case of homotopy finiteness, hence some of the results we give are special cases of Wall's and the proofs are altered accordingly.

2.1 Attaching cells

As we are ultimately interested in homotopy equivalences between CW complexes, and these are given from weak homotopy equivalences by Whitehead's theorem, a key construction will be to alter maps of CW complexes so that they induce isomorphisms on homotopy groups up to progressively higher dimensions. Then induction will allow us to turn this into a method for constructing homotopy equivalences. The construction is taken from the "(iii) $\implies$ (i)" direction in the proof of Theorem A in Wall [Wal65].

The construction is given by following several lemmas.

Lemma 2.6. If Y is a CW complex and Z is obtained from Y by attaching $(i+1)$ -cells, then $\pi_i(Z, Y) = 0$.

Proof. Since Z only differs from Y by $(i+1)$ -cells, they share an i -skeleton Y^i . Then the portion of the homotopy exact sequence of the triple (Z, Y, Y^i)

$$\pi_i(Y, Y^i) \rightarrow \pi_i(Z, Y^i) \rightarrow \pi_i(Z, Y) \rightarrow \pi_{i-1}(Y, Y^i)$$

has all other terms equal to zero since (Y, Y^i) and (Z, Y^i) are i -connected by Proposition 1.26, hence also $\pi_i(Z, Y) = 0$. ■

Lemma 2.7 ([Wal65, p. 59]). Let X and Y be CW complexes and $f : Y \rightarrow X$ be an i -connected map. Then there exists an $(i+1)$ -connected map $g : Z \rightarrow X$ such that Z is obtained from Y by attaching $(i+1)$ -cells and $g|_Y = f$. The number of cells attached can be taken equal to the number of generators of $\pi_{i+1}(f)$ as a $\mathbb{Z}[\pi_1(X)]$ -module if $i \geq 2$.

Proof. Since we can replace X by the mapping cylinder M_f of f , we can assume that f is an inclusion. Now an element of $\pi_{i+1}(f) = \pi_{i+1}(X, Y)$ is represented by a map $\alpha : (D^{i+1}, S^i) \rightarrow (X, Y)$, which can be taken to be cellular by the cellular approximation theorem. Then we can use $\alpha|_{S^i}$ to attach an $(i+1)$ -cell to Y , giving a new CW complex Y' . Note here the importance of α being cellular: it is only because of this that the cell attaches to Y^i and hence Y' is a CW complex. We can extend f to $f' : Y' \rightarrow X$ by letting it equal α on the $(i+1)$ -cell. We can do this for more than one α at a time, using a union of the different attaching maps as in the definition of a CW complex. In particular, we can do it for every element of $\pi_{i+1}(f)$. If $i \geq 2$ however, we have that f induces an isomorphism of fundamental groups, hence $\pi_{i+1}(f)$ takes on the structure of a $\mathbb{Z}[\pi_1(X)]$ -module, and we can instead use a set of generators of $\pi_{i+1}(f)$. Perform the attachment and call the resulting complex Z with extended map $g : Z \rightarrow X$. Clearly $g|_Y = f$; it remains to check that g is $(i+1)$ -connected.

For this, consider the mapping cylinder M_g of g . It contains as a subspace the mapping cylinder M_f of f , and from this we can easily see that $\pi_{i+1}(M_g, Y) \cong \pi_{i+1}(f)$. We can either see this directly from the definitions, or by considering the homotopy exact sequence of the triple (M_g, M_f, Y) .

Now consider the following portion of the homotopy exact sequence of the triple (M_g, Z, Y) :

$$\pi_{i+1}(Z, Y) \xrightarrow{\phi} \pi_{i+1}(f) \rightarrow \pi_{i+1}(g) \rightarrow \pi_i(Z, Y) = 0.$$

The last term is zero by Lemma 2.6. If we show that ϕ is surjective, then $\pi_{i+1}(g) = 0$ and g is $(i+1)$ -connected as required, so we are done.

To see that ϕ is surjective, recall that it is given by inclusion, i.e. it takes a map $(D^{i+1}, S^i) \rightarrow (Z, Y)$ to its composition with $g : (Z, Y) \rightarrow (X, Y)$. In particular, all the maps we used to construct Z will be in the image of ϕ . But we selected a generating set for $\pi_{i+1}(f)$ (or the whole of $\pi_{i+1}(f)$ if $i = 0, 1$), so ϕ is indeed surjective. ■

Lemma 2.8 ([Wal65, Lemma 1.2]). Let X and Y be CW complexes and $f : Y \rightarrow X$ be an i -connected map. Then there exists a homotopy equivalence $g : Z \rightarrow X$ such that Z is obtained from Y by attaching cells of dimensions $(i+1)$ and above and $g|_Y = f$.

Proof. By repeatedly using Lemma 2.7, we obtain Z from Y by attaching cells of dimensions $i+1$ and above, and $g : Z \rightarrow X$ n -connected for all n . Hence g is a weak homotopy equivalence, and thus a homotopy equivalence by Whitehead's theorem. ■

Remark 2.9. This lemma means that, as long as we are only interested in CW complexes up to homotopy equivalence, we can regard i -connected maps as inclusions of a subcomplex with common i -skeleton.

2.2 Equivalence to CW complex of finite type

Now we turn our attention to the matter at hand. In this section we aim to prove the following theorem as a first step toward Theorem 2.1:

Theorem 2.10. A finitely dominated CW complex X is homotopy equivalent to a CW complex of finite type.

This will then immediately imply (by Corollary 1.60):

Corollary 2.11. If a CW complex X is finitely dominated, then $C_*(\tilde{X})$ is chain homotopy equivalent to a free chain complex of finite type.

For the proof of Theorem 2.10 we will follow the “(ii) $\implies$ (i)” direction in the proof of Theorem A in Wall [Wal65].

Before we start, we will need a purely group theoretic lemma (proof reproduced from Wall's almost exactly):

Lemma 2.12 ([Wal65, Lemma 1.3]). Let G be a finitely presented group and $r : G \rightarrow H$ and $s : H \rightarrow G$ be homomorphisms such that $r \circ s = \text{id}_H$. Then H is finitely presented.

Proof. First note that the condition $r \circ s = \text{id}_H$ gives that r is surjective and s is injective. Now let $\{g_1, \dots, g_n\}$ be a finite generating set for G . Then, as r is surjective, we have that $\{r(g_1), \dots, r(g_n)\}$ is a finite generating set for H . Consider the elements of G given by $g_i^{-1}s(r(g_i))$, and let K be the smallest normal subgroup of G containing these elements. Clearly $K \leq \ker r$, so we have surjective homomorphisms $q : G \rightarrow G/K$ and $p : G/K \rightarrow H$ such that $p \circ q = r$. This then means that $p \circ q \circ s = \text{id}_H$, so $q \circ s : H \rightarrow G/K$ is injective. On the other hand, by the fact that $q(g_i) = q(s(r(g_i)))$ in G/K by construction, and since the $q(g_i)$ generate G/K , we get that G/K is generated by $q(s(r(g_i)))$, so $q \circ s$ is also surjective, hence an isomorphism. Now G/K is finitely presented since we can obtain a presentation by adding the relations $g_i^{-1}s(r(g_i))$ to a finite presentation of G , and we are done. $\blacksquare$

Corollary 2.13. For X a finitely dominated CW complex, $\pi_1(X)$ is a finitely presented group.

Motivated by Lemma 2.12, we now take a moment to make an aside and prove a similar lemma for R -modules. This result is not in Wall's papers, and in fact, though it is quite elementary, the author is not aware of a proof of it in the literature:

Lemma 2.14. Let M be a finitely presented R -module and $r : M \rightarrow N$ and $s : N \rightarrow M$ be R -module homomorphisms such that $r \circ s = \text{id}_N$. Then N is a finitely presented R -module.

Proof. Clearly N is finitely generated. Next note that r is a split surjection, hence we have $M = N \oplus N'$ for some finitely generated N' . Since M is finitely presented, we have a surjection $p : R^n \rightarrow M$ such that $\ker p$ is finitely generated for some n . Then $r \circ p : R^n \rightarrow N$ is a surjection, and we need that $\ker(r \circ p)$ is finitely generated. This is given by the short exact sequence

$$0 \rightarrow \ker p \rightarrow \ker(r \circ p) \xrightarrow{p} N' \rightarrow 0. \quad \blacksquare$$

Now let us consider how we will prove Theorem 2.10. We can use the inductive construction from Section 2.1 (Lemmas 2.7 and 2.8): to do this, we will need a base case (i.e. a finite complex Y and an i -connected map $f : Y \rightarrow X$ for some $i \geq 2$), and then we will need to ensure that at each inductive step $\pi_{i+1}(f)$ is a finitely generated $\mathbb{Z}[\pi_1(X)]$ -module.

Focusing on the second issue, recall from Remark 1.11 that f induces a map $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ and we can see by the Hurewicz theorem that $\pi_{i+1}(f) \cong H_{i+1}(\tilde{f})$. So it would be enough to show this second module to be finitely generated. Were it true that for K a finite CW complex $H_n(\tilde{K})$ are finitely presented, we could make the following attempt at a proof, for any i -connected map $f : Y \rightarrow X$:

Proof attempt. Note that $H_i(\tilde{Y})$ is finitely presented since Y is finite, and also $H_{i+1}(\tilde{X})$ and $H_i(\tilde{X})$ are finitely presented by Lemma 2.14. Then the portion of the homology exact sequence of $\tilde{f}$

$$H_{i+1}(\tilde{X}) \xrightarrow{q} H_{i+1}(\tilde{f}) \xrightarrow{\partial} H_i(\tilde{Y}) \xrightarrow{\tilde{f}_*} H_i(\tilde{X}) \rightarrow 0$$

gives first that $\text{im } \partial \cong \ker \tilde{f}_*$ is finitely generated (since $H_i(\tilde{X})$ is finitely presented) and then that the short exact sequence

$$0 \rightarrow \text{im } q \rightarrow H_{i+1}(\tilde{f}) \rightarrow \text{im } \partial \rightarrow 0$$

has the first and last terms finitely generated, hence also the middle term. ■

However, as we discussed in Remark 1.51, the $H_n(\tilde{K})$ for finite K need not even be finitely generated: Stallings [Sta63, Corollary 2] constructs a finite CW complex K with $\pi_2(K) \cong \pi_2(\tilde{K}) \cong H_2(\tilde{K})$ (by the Hurewicz theorem) not finitely generated as a $\mathbb{Z}[\pi_1(K)]$ -module. Thus the above attempt fails, and we will need some further property of the map f to be able to conclude that $H_{i+1}(\tilde{f})$ is finitely generated. The property in question will turn out to be that f is a domination, i.e. that there exists a $g : X \rightarrow Y$ with $f \circ g \simeq \text{id}_X$, and we will see how this arises in the proof of Theorem 2.10. But first we give some lemmas which will be helpful in showing that $H_{i+1}(\tilde{f})$ is finitely generated:

Lemma 2.15. For a CW complex X with finitely many i -cells, the homology module $H_i(\tilde{X}, \tilde{X}^{i-1})$ is finitely generated.

Proof. Recall that $H_i(\tilde{X}^i, \tilde{X}^{i-1})$ is a free module with basis the i -cells of X by Proposition 1.50. Recall also that the pair $(\tilde{X}, \tilde{X}^i)$ is i -connected by Proposition 1.26. Then the result follows from the homology exact sequence of the triple $(\tilde{X}, \tilde{X}^i, \tilde{X}^{i-1})$:

$$H_i(\tilde{X}^i, \tilde{X}^{i-1}) \rightarrow H_i(\tilde{X}, \tilde{X}^{i-1}) \rightarrow 0. \quad \blacksquare$$

The next lemma is used by Wall [Wal65, p. 60] with a one-sentence justification; we write out the proof:

Lemma 2.16. Let X be a CW complex with dominating complex K , and let f and g be the maps with $f \circ g \simeq \text{id}_X$. Suppose that f induces an isomorphism on fundamental groups. Then $H_{i+1}(\tilde{f}) \cong H_i(\tilde{g})$ as $\mathbb{Z}[\pi_1(X)]$ -modules for all $i \geq 0$.

Proof. We have induced $\mathbb{Z}[\pi_1(X)]$ -module homomorphisms $\tilde{g}_* : H_n(\tilde{X}) \rightarrow H_n(\tilde{K})$ and $\tilde{f}_* : H_n(\tilde{K}) \rightarrow H_n(\tilde{X})$ such that $\tilde{f}_* \circ \tilde{g}_* = \text{id}_{H_n(\tilde{X})}$ for all n . This gives that $\tilde{f}_*$ is surjective for all n and hence by the homology exact sequence of $\tilde{f}$ we get a short exact sequence

$$0 \rightarrow H_{i+1}(\tilde{f}) \rightarrow H_i(\tilde{K}) \xrightarrow{\tilde{f}_*} H_i(\tilde{X}) \rightarrow 0$$

which is split by $\tilde{g}_*$, giving $H_{i+1}(\tilde{f}) \cong \text{coker } \tilde{g}_*$ (due to $H_i(\tilde{K}) \cong \text{im } \tilde{g}_* \oplus \ker \tilde{f}_*$ and the second isomorphism theorem for modules).

But similarly $\tilde{g}_*$ is injective for all n and the homology exact sequence of $\tilde{g}$ gives a short exact sequence

$$0 \rightarrow H_i(\tilde{X}) \xrightarrow{\tilde{g}_*} H_i(\tilde{K}) \rightarrow H_i(\tilde{g}) \rightarrow 0$$

and hence $\text{coker } \tilde{g}_* \cong H_i(\tilde{g})$. ■

We are now ready to prove Theorem 2.10 in full. It is a special case of Theorem A in Wall [Wal65], and we reproduce the proof with added detail:

Theorem 2.10. A finitely dominated CW complex X is homotopy equivalent to a CW complex of finite type.

Proof. As mentioned, we use the inductive construction (Lemmas 2.7 and 2.8) to prove this theorem, and we need a base case (a finite complex Y and an i -connected map $f : Y \rightarrow X$ for some $i \geq 2$) and to ensure that at each inductive step $\pi_{i+1}(f)$ is finitely generated.

Let's deal with the second issue first. Assuming that $f : Y \rightarrow X$ is i -connected with Y finite, we need to show that $\pi_{i+1}(f)$ is finitely generated. But for $i \geq 2$

$$\pi_{i+1}(f) \cong \pi_{i+1}(\tilde{f}) \cong H_{i+1}(\tilde{f})$$

by the Hurewicz theorem. So in fact it is enough to show that the homology module $H_{i+1}(\tilde{f})$ is finitely generated, and then ensure the base case starts at $i \geq 2$.

Now recall that by Lemma 2.8 we can consider f to be the inclusion of Y as a subcomplex of X with common i -skeleton $X^i = Y^i$, and therefore also common $(i-1)$ -skeleton $X^{i-1} = Y^{i-1}$. The same is true for the universal covers. Now let $g : X \rightarrow Y$ be any map and let $M_{\tilde{g}}$ be the mapping cylinder of the induced map $\tilde{g} : \tilde{X} \rightarrow \tilde{Y}$; then, noting that $H_i(M_{\tilde{g}}, \tilde{X}^{i-1}) \cong H_i(\tilde{Y}, \tilde{X}^{i-1})$, consider the homology exact sequence of the triple $(M_{\tilde{g}}, \tilde{X}, \tilde{X}^{i-1})$

$$H_i(\tilde{Y}, \tilde{X}^{i-1}) \rightarrow H_i(\tilde{g}) \rightarrow H_{i-1}(\tilde{X}, \tilde{X}^{i-1}) = 0$$

where the last term is zero since $(\tilde{X}, \tilde{X}^{i-1})$ is $(i-1)$ -connected. Therefore $H_i(\tilde{g})$ is a quotient of $H_i(\tilde{Y}, \tilde{X}^{i-1})$, and this we know to be finitely generated by Lemma 2.15.

This means that we would be done with the inductive step if we could establish an isomorphism $H_{i+1}(\tilde{f}) \cong H_i(\tilde{g})$. Such an isomorphism is given by Lemma 2.16, but we need the additional assumption that f is a domination with homotopy right-inverse g . This condition is easily seen to be preserved by our construction of attaching cells to Y via Lemma 2.7, so all we need to do is make sure it is satisfied by our base case.

Thus for a base case, we might first consider the finite dominating complex K of X . This comes with a 1-connected map $f' : K \rightarrow X$. However it doesn't fully work, since we need a 2-connected map, and also in general we won't have $\pi_1(K) = \pi_1(X)$. We would like to amend this by attaching cells to K to get a finite complex with the correct fundamental group, and then we'll check the resulting map is 2-connected.

By Lemma 2.12 we can pick appropriate presentations for $\pi_1(X)$ and $\pi_1(K)$ which only differ by finitely many extra relations for $\pi_1(X)$, say α_i . By the homotopy exact sequence

$$\pi_2(f') \rightarrow \pi_1(K) \rightarrow \pi_1(X)$$

the α_i are all in the image of the first map, so they come from $\alpha'_i \in \pi_2(f')$, and we can use the α'_i to attach 2-cells to K and extend f' over these cells like in Lemma 2.7. This gives a finite complex Y with map $f : Y \rightarrow X$ inducing an isomorphism on fundamental groups. Also $f|_K = f'$, so Y still dominates X with the same homotopy right-inverse as before. This means that f induces a surjection on π_2 , so by inspection of the homotopy exact sequence

$$\pi_2(Y) \rightarrow \pi_2(X) \rightarrow \pi_2(f) \rightarrow \pi_1(Y) \rightarrow \pi_1(X)$$

we see that $\pi_2(f) = 0$, and hence f is 2-connected. This completes the base case, and the proof of the theorem. ■

2.3 $C_*(\tilde{X})$ is finitely dominated

In this section we finish the proof of Theorem 2.1, by taking the free chain complex of finite type given by Corollary 2.11 and truncating it suitably into a finite projective complex. This proof is taken from Wall's second paper [Wal66, Theorem 6], with added detail.

Theorem 2.1 ([Wal65, Theorem 6]). If a CW complex X is finitely dominated, then $C_*(\tilde{X})$ is a finitely dominated chain complex.

Proof. Let n be the dimension of the dominating complex. By Corollary 2.11, we can assume $F_* \simeq C_*(\tilde{X})$ to be a free chain complex of finite type, and it is enough to show that F_* is chain equivalent to a finite projective complex. By our comments on cohomology at the end of Section 1.4, we have that $H^{n+1}(\tilde{X}; B) = 0$ for any $\mathbb{Z}[\pi_1(X)]$ -module B . Hence also $H^{n+1}(F_*, B) = 0$; in particular, choose B to be $\text{im } d_{n+1}$, where $d_{n+1} : F_{n+1} \rightarrow F_n$ is the boundary map of F_* . That is, we have that the sequence

$$\text{Hom}_{\mathbb{Z}[\pi_1(X)]}(F_n, \text{im } d_{n+1}) \xrightarrow{\partial_n} \text{Hom}_{\mathbb{Z}[\pi_1(X)]}(F_{n+1}, \text{im } d_{n+1}) \xrightarrow{\partial_{n+1}} \text{Hom}_{\mathbb{Z}[\pi_1(X)]}(F_{n+2}, \text{im } d_{n+1}),$$

is exact in the middle.

Decompose the map $d_{n+1} = \iota \circ d'_{n+1}$, where $\iota : \text{im } d_{n+1} \rightarrow F_n$ is the inclusion and $d'_{n+1} : F_{n+1} \rightarrow \text{im } d_{n+1}$ is the obvious map. Then the map d'_{n+1} is an element of $\text{Hom}_{\mathbb{Z}[\pi_1(X)]}(F_{n+1}, \text{im } d_{n+1})$ which is in $\ker \partial_{n+1}$ by the chain complex property of the boundary map. By exactness in the middle of the above sequence, we get $d'_{n+1} \in \text{im } \partial_n$, hence there exists $r : F_n \rightarrow \text{im } d_{n+1}$ with $d'_{n+1} = r \circ d_{n+1} = r \circ \iota \circ d'_{n+1}$. But d'_{n+1} is surjective, so we can cancel it, giving $\text{id}_{\text{im } d_{n+1}} = r \circ \iota$. Therefore r splits the short exact sequence

$$0 \rightarrow \text{im } d_{n+1} \xrightarrow{\iota} F_n \rightarrow \text{coker } \iota \rightarrow 0,$$

yielding that $\text{im } d_{n+1}$ is a direct summand of F_n .

Let P_n be the finitely generated projective module such that $P_n \oplus \text{im } d_{n+1} = F_n$ and define the following chain map

$$\begin{array}{ccccccc} \cdots & \longrightarrow & F_{n+1} & \longrightarrow & F_n & \longrightarrow & F_{n-1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow q & & \downarrow \text{id} \\ \cdots & \longrightarrow & 0 & \longrightarrow & P_n & \longrightarrow & F_{n-1} \longrightarrow \cdots \end{array}$$

with q the quotient map and identity maps at levels below n . This chain map clearly induces isomorphisms on homology in levels other than n , and at level n this is true because of the correspondence result about submodules of quotient modules. As the complexes consist are projective, the chain map is a chain homotopy equivalence by Theorem 1.59, and we are done. $\blacksquare$

2.4 Euler class is a homotopy invariant

Our task in this section is to take care of the final point in making sure the Wall finiteness obstruction is well defined, which is Theorem 2.3. We work with chain complexes of R -modules for an arbitrary ring R .

Before the proof of the theorem, we need the simple fact that the Euler class of an exact complex vanishes (this is a special case of Proposition 1.7.10 in Rosenberg [Ros94], with similar proof):

Lemma 2.17. If P_* is an exact finite projective chain complex, then $\chi(P_*) = 0$.

Proof. We claim that $P_i = \text{im } d_{i+1} \oplus \text{im } d_i$ for all i . It's trivial that this is the case for $i = 0$, and we use this as a base case for induction.

Now suppose $P_{i-1} = \text{im } d_i \oplus \text{im } d_{i-1}$. Then $\text{im } d_i$ is projective. Exactness at level i of P_* gives a short exact sequence

$$0 \rightarrow \text{im } d_{i+1} = \ker d_i \rightarrow P_i \xrightarrow{d_i} \text{im } d_i \rightarrow 0$$

which splits since $\text{im } d_i$ is projective, proving the claim.

Now (remembering that the sums have only finitely many nonzero terms) we have that $\chi(P_*) = \sum_{i=0}^{\infty} (-1)^i [P_i] = \sum_{i=0}^{\infty} (-1)^i ([\text{im } d_{i+1}] + [\text{im } d_i]) = 0$. $\blacksquare$

Now we can show the homotopy invariance of the Euler class. We follow closely the proof in Rosenberg [Ros94, Theorem 1.7.11]; a different proof is given by Ranicki [Ran85, Proposition 1.2]:

Theorem 2.3 ([Ros94, Corollary 1.7.11]). If P_* and P'_* are two homotopy equivalent finite projective chain complexes, then $\chi(P_*) = \chi(P'_*)$.

Proof. If $f : P_* \rightarrow P'_*$ is the homotopy equivalence, we know by Theorem 1.59 that M_*^f is an exact complex. Since P_* and P'_* are finite and projective, so is M_*^f , hence we can use Lemma 2.17 to find that it has vanishing Euler class. We have:

$$\begin{aligned} 0 = \chi(M_*^f) &= \sum_{i=0}^{\infty} (-1)^i [P'_i \oplus P_{i-1}] \\ &= \left(\sum_{i=0}^{\infty} (-1)^i [P'_i] \right) - \left(\sum_{i=0}^{\infty} (-1)^i [P_i] \right) \\ &= \chi(P'_*) - \chi(P_*), \end{aligned}$$

where the third equality is allowed because the sums have only finitely many nonzero terms. $\blacksquare$

Chapter 3

Vanishing of the obstruction

In this section we complete the proof of the main result:

Theorem 3.1 ([Wal65, Theorem F]). A CW complex X is homotopy finite if and only if it is finitely dominated and $\omega(X) = 0$.

To do this, we take a slightly different route from Wall's, to make use of an algebraic result of Ranicki [Ran85]. Section 3.1 contains an overview of the proof and where it differs from Wall's. Section 3.2 then gives Ranicki's result on the vanishing of the reduced Euler class of a finitely dominated complex, and Section 3.3 finishes the proof by geometrically realising a finite free chain complex as $C_*(\tilde{Y})$ for a CW complex Y , borrowing from the book by Johnson [Joh12] on the way.

3.1 High-level overview

First we make a convenient definition:

Definition 3.2. Let G be a group and let F_* be a free chain complex of $\mathbb{Z}[G]$ -modules. Suppose there exists a CW complex Y with $\pi_1(Y) \cong G$ and $C_*(\tilde{Y}) \simeq F_*$. Then we say that Y *realises* F_* .

Recall that the goal is to prove the main theorem:

Theorem 3.1 ([Wal65, Theorem F]). A CW complex X is homotopy finite if and only if it is finitely dominated and $\omega(X) = 0$.

If X is homotopy finite, it is clearly finitely dominated and $C_*(\tilde{X})$ is a chain equivalent to a finite free complex F_* by Corollary 1.60, hence $\omega(X) = \tilde{\chi}(F_*) = 0$ by Theorem 2.3. So the forward direction is easy; the hard direction is thus to start with a finitely dominated complex X and prove that it is homotopy finite, i.e. we are looking for a finite complex Y and a homotopy equivalence $f : X \rightarrow Y$.

So how do we construct Y and f ? Let's look at the cellular complexes of X and Y :

- Y is finite, so $C_*(\tilde{Y})$ is a finite free chain complex;
- X is finitely dominated, so $C_*(\tilde{X})$ is a finitely dominated chain complex (by Theorem 2.1) with $\tilde{\chi}(C_*(\tilde{X})) = 0$.

Thus f induces a chain equivalence $f_* : C_*(\tilde{Y}) \rightarrow C_*(\tilde{X})$ for a finite free chain complex $C_*(\tilde{Y})$. So we could break the problem down into the following steps:

1. Use the vanishing reduced Euler class of $C_*(\tilde{X})$ to find a chain equivalence $f_* : F_* \rightarrow C_*(\tilde{X})$ where F_* is some finite free chain complex.
2. Construct a finite CW complex Y realising F_* and a homotopy equivalence $f : Y \rightarrow X$ inducing f_* .

The two steps are handled in Sections 3.2 and 3.3 respectively.

Wall's original proof [Wal65, Theorem F] is more direct, employing the inductive process from the proof of Theorem 2.10 and showing that the vanishing of the obstruction is sufficient for this process to terminate after finitely many steps. Wall's proof is also better in the sense that it gives the dimension of the resulting Y —although it is also not hard to read this off from the proof given here as well. We will not be concerned with this for now, but we will come back to it in Chapter 4.

The reason we give this alternate proof is to highlight two points: firstly, the parallel with the algebraic result of Ranicki offers some insight into just how strong of an analogy exists between geometry and algebra, CW complexes and chain complexes, and how helpful this analogy can turn out to be in proofs; secondly, the realisation problem tackled in Section 3.3 is really hardest in lowest dimensions, and this speaks to a connection with the unsolved D2 problem which we will touch upon in Chapter 4.

Ultimately, the proof given here is “the same” as Wall's, except it reveals more of what is going on beneath the surface.

3.2 Vanishing of the reduced Euler class

In this section we take care of the first step of the proof as outlined in Section 3.1. The following theorem is due to Ranicki [Ran85]; we follow closely the proof in Rosenberg [Ros94, Theorem 1.7.12].

Theorem 3.3. A finitely dominated chain complex C_* is homotopy equivalent to a finite free chain complex if and only if $\tilde{\chi}(C_*) = 0$.

Proof. Suppose C_* is homotopy equivalent to a finite free chain complex F_* . Then we have that $[F_i] = 0 \in \tilde{K}_0(R)$ for all i since F_i is free, and hence also $\tilde{\chi}(F_*) = 0$, and hence $\tilde{\chi}(C_*) = 0$ by Theorem 2.3.

Now let P_* be a finite projective complex equivalent to C_* ; then $\tilde{\chi}(P_*) = 0$ by Theorem 2.3. Let n be the dimension of P_* . Since P_0 is projective, there exists another projective module Q_0 such that $P_0 \oplus Q_0 = F_0$ with F_0 free (note that all three modules are finitely generated). We can then alter P_* into an equivalent complex by taking the direct sum $P_* \oplus (Q_0)_*^1$ with the elementary complex $(Q_0)_*^1$. (Recall that taking direct sums with elementary complexes preserves chain homotopy type; see discussion after Definition 1.55.) If $n = 0$, Q_0 can be taken to be free (since $[P_0] = 0 \in \tilde{K}_0(R)$ and hence P_0 is stably free) and we are done. Otherwise, the resulting complex is

$$P'_* = P_n \rightarrow \cdots \rightarrow P_2 \rightarrow P_1 \oplus Q_0 \rightarrow F_0.$$

Now since $P_1 \oplus Q_0$ is projective, we can choose another projective module Q_1 such that $P_1 \oplus Q_0 \oplus Q_1 = F_1$ with F_1 free (again all modules are finitely generated). Then we get

$$P'_* \oplus (Q_1)_*^2 = P_n \rightarrow \cdots \rightarrow P_2 \oplus Q_1 \rightarrow F_1 \rightarrow F_0.$$

We can continue in this manner, replacing P_* at step i with a homotopy equivalent chain complex whose modules up to dimension $(i - 1)$ are finitely generated and free. After step n we have produced the complex

$$P''_* = P_n \oplus Q_{n-1} \rightarrow F_{n-1} \rightarrow \cdots \rightarrow F_0,$$

and since this is homotopy equivalent to P_* it has vanishing reduced Euler class (again using Theorem 2.3). But $[F_i] = 0 \in \tilde{K}_0(R)$ for all i because the F_i are free, hence we have that $[P_n \oplus Q_{n-1}] = \tilde{\chi}(P''_*) = 0 \in \tilde{K}_0(R)$, and so $P_n \oplus Q_{n-1}$ is stably free. Then we can choose a finitely generated free module F' such that $P_n \oplus Q_{n-1} \oplus F' = F_n$ is finitely generated and free, and we can take one last direct sum to produce the desired finite free complex equivalent to P_* :

$$P''_* \oplus (F')_*^n = F_n \rightarrow F'_{n-1} \rightarrow \cdots \rightarrow F_0,$$

where $F'_{n-1} = F_{n-1} \oplus F'$ is still finitely generated and free. ■

The following immediately corollary then achieves the first step of the proof in Section 3.1:

Corollary 3.4. If X is a finitely dominated CW complex with $\omega(X) = 0$, then $C_*(\tilde{X})$ is chain equivalent to a finite free chain complex.

3.3 Geometric realisation

We are now at the final stage where, having started with a finitely dominated CW complex X with $\omega(X) = 0$, using Corollary 3.4 we can turn the $C_*(\tilde{X})$ into a finite free chain complex, say F_* . The last step now is to use F_* to construct a finite CW complex Y realising F_* and to prove that Y is homotopy equivalent to X . We will do this by induction as in the proof of Theorem 2.10, but the base case is even more troublesome now, since a finite 2-dimensional complex Y' with the correct fundamental group and a 2-connected map is not enough: we also need Y' to realise the 2-skeleton of F_* . We will separate this base case out in a lemma, taken from Johnson [Joh12], and then we will finish the proof following Wall again.

First, we need a very well known lemma from homological algebra:

Lemma 3.5 (Schanuel's Lemma [Joh12, Proposition 5.10]). Let

$$\begin{aligned} 0 \rightarrow Q_1 \xrightarrow{p_1} P_1 \xrightarrow{q_1} M \rightarrow 0 \\ 0 \rightarrow Q_2 \xrightarrow{p_2} P_2 \xrightarrow{q_2} M' \rightarrow 0 \end{aligned}$$

be short exact sequences with P_1 and P_2 projective, and let $h : M \rightarrow M'$ be an isomorphism. Then there exists a chain isomorphism

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q_1 \oplus P_2 & \xrightarrow{p_1 \oplus \text{id}} & P_1 \oplus P_2 & \xrightarrow{(q_1, 0)} & M \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow \wr & & \downarrow h \\ 0 & \longrightarrow & Q_2 \oplus P_1 & \xrightarrow{p_2 \oplus \text{id}} & P_2 \oplus P_1 & \xrightarrow{(q_2, 0)} & M' \longrightarrow 0. \end{array}$$

Proof. Let N be the kernel of the map $P_1 \oplus P_2 \xrightarrow{(q_1, h^{-1}q_2)} M$. Then take the composition $N \rightarrow P_1 \oplus P_2 \rightarrow P_1$, where the first map is the inclusion and the second is the projection, and check that this is a surjection with kernel isomorphic to Q_2 . The surjection is split since P_1 is projective, hence $N \cong Q_2 \oplus P_1$. Similarly, $N \cong Q_1 \oplus P_2$. Then fit this isomorphism into a diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q_1 \oplus P_2 & \xrightarrow{p_1 \oplus \text{id}} & P_1 \oplus P_2 & \xrightarrow{(q_1, 0)} & M \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow & & \downarrow h \\ 0 & \longrightarrow & Q_2 \oplus P_1 & \xrightarrow{p_2 \oplus \text{id}} & P_2 \oplus P_1 & \xrightarrow{(q_2, 0)} & M' \longrightarrow 0, \end{array}$$

where the middle map is given by the projectivity of $P_1 \oplus P_2$. We can check that the diagram is commutative by unfurling the definition of the isomorphism on the left; then the middle map is also an isomorphism by the five lemma. $\blacksquare$

Next is the realisation lemma which will be our base case. The proof is taken from Johnson [Joh12]: it is contained in the proof of Theorem 8.26 but is essentially a direct application of a special case of Theorem 8.9.

Lemma 3.6. Let G be a finitely presented group and $F_* = (F_2 \xrightarrow{f_2} F_1 \xrightarrow{f_1} F_0)$ be a finite free complex of $\mathbb{Z}[G]$ -modules with $H_1(F_*) = 0$ and $H_0(F_*) \cong \mathbb{Z}$ with the trivial $\mathbb{Z}[G]$ -action. Then there exists a finite 2-dimensional CW complex Y realising $(F_2 \oplus F' \xrightarrow{(f_2, 0)} F_1 \xrightarrow{f_1} F_0)$ for some finitely generated free module F' .

Remark 3.7. The hypotheses about the homology of F_* are necessary: we cannot possibly hope to be able to realise F_* otherwise, for $C_*(\tilde{Y})$ will have the specified homology since $\tilde{Y}$ is simply-connected.

Proof. Let $\langle S|R \rangle$ be a finite presentation for G , and let K be the presentation complex for G with this presentation. Recall that K is a 2-dimensional connected CW complex with one 0-cell, $|S|$ 1-cells, and $|R|$ 2-cells. Hence we have that $C_*(\tilde{K}) = (E_2 \xrightarrow{g_2} E_1 \xrightarrow{g_1} E_0 = \mathbb{Z}[G])$ where all three modules are finitely generated and free. We can attach any (finite) number of 2-cells to the basepoint of K , resulting in a CW complex Y with $C_*(\tilde{Y}) = (E_2 \oplus E' \xrightarrow{(g_2, 0)} E_1 \xrightarrow{g_1} E_0)$ for some finitely generated free module E' . Hence it's enough to show that this chain complex is chain homotopy equivalent to a complex of the required form.

Start off by considering the two short exact sequences

$$\begin{aligned} 0 &\rightarrow \text{im } g_1 \rightarrow E_0 \xrightarrow{g_1} \mathbb{Z} \rightarrow 0 \\ 0 &\rightarrow \text{im } f_1 \rightarrow F_0 \xrightarrow{q_2} \mathbb{Z} \rightarrow 0 \end{aligned}$$

given by the definitions of $H_0(\tilde{K})$ and $H_0(F_*)$ respectively. By Schanuel's lemma, these give a chain isomorphism

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{im } g_1 \oplus F_0 & \longrightarrow & E_0 \oplus F_0 & \xrightarrow{(q_1, 0)} & \mathbb{Z} \longrightarrow 0 \\ & & \downarrow \phi & & \downarrow \wr & & \downarrow \text{id} \\ 0 & \longrightarrow & \text{im } f_1 \oplus E_0 & \longrightarrow & F_0 \oplus E_0 & \xrightarrow{(q_2, 0)} & \mathbb{Z} \longrightarrow 0. \end{array} \tag{3.1}$$

Next, since F_* and $C_*(\tilde{K})$ are exact in the middle, we have the short exact sequences

$$0 \rightarrow \operatorname{im} g_2 \rightarrow E_1 \xrightarrow{g_1} \operatorname{im} g_1 \rightarrow 0$$

$$0 \rightarrow \operatorname{im} f_2 \rightarrow F_1 \xrightarrow{f_1} \operatorname{im} f_1 \rightarrow 0$$

and they remain exact after direct sum with elementary complexes $(F_0)_*^1$ and $(E_0)_*^1$ respectively:

$$0 \rightarrow \operatorname{im} g_2 \rightarrow E_1 \oplus F_0 \xrightarrow{g_1 \oplus \operatorname{id}} \operatorname{im} g_1 \oplus F_0 \rightarrow 0$$

$$0 \rightarrow \operatorname{im} f_2 \rightarrow F_1 \oplus E_0 \xrightarrow{f_1 \oplus \operatorname{id}} \operatorname{im} f_1 \oplus E_0 \rightarrow 0.$$

Now using the isomorphism ϕ from (3.1), we can apply Schanuel's Lemma again to get the chain isomorphism

$$\begin{array}{ccccccc} 0 & \longrightarrow & \operatorname{im} g_2 \oplus D & \longrightarrow & (E_1 \oplus F_0) \oplus D & \xrightarrow{(g_1 \oplus \operatorname{id}, 0)} & \operatorname{im} g_1 \oplus F_0 \longrightarrow 0 \\ & & \downarrow \psi & & \downarrow \wr & & \downarrow \phi \\ 0 & \longrightarrow & \operatorname{im} f_2 \oplus D' & \longrightarrow & (F_1 \oplus E_0) \oplus D' & \xrightarrow{(f_1 \oplus \operatorname{id}, 0)} & \operatorname{im} f_1 \oplus E_0 \longrightarrow 0, \end{array}$$

where $D = F_1 \oplus E_0$ and $D' = E_1 \oplus F_0$. Now we can glue this chain isomorphism to (3.1) to get a longer chain isomorphism of exact sequences

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & \operatorname{im} g_2 \oplus D & \longrightarrow & (E_1 \oplus F_0) \oplus D & \xrightarrow{(g_1 \oplus \operatorname{id}, 0)} & E_0 \oplus F_0 & \xrightarrow{(q_1, 0)} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \psi & & \downarrow \wr & & \downarrow \wr & & \downarrow \operatorname{id} & & \\ 0 & \longrightarrow & \operatorname{im} f_2 \oplus D' & \longrightarrow & (F_1 \oplus E_0) \oplus D' & \xrightarrow{(f_1 \oplus \operatorname{id}, 0)} & F_0 \oplus E_0 & \xrightarrow{(q_2, 0)} & \mathbb{Z} & \longrightarrow & 0. \end{array}$$

Notice that we can “cancel” the elementary complexes $(F_0)_*^2$ and $(E_0)_*^2$ from the rows of this diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \operatorname{im} g_2 \oplus D & \longrightarrow & E_1 \oplus D & \xrightarrow{(g_1, 0)} & E_0 \xrightarrow{q_1} \mathbb{Z} \longrightarrow 0 \\ & & \downarrow \psi & & \downarrow & & \downarrow & \downarrow \operatorname{id} \\ 0 & \longrightarrow & \operatorname{im} f_2 \oplus D' & \longrightarrow & F_1 \oplus D' & \xrightarrow{(f_1, 0)} & F_0 \xrightarrow{q_2} \mathbb{Z} \longrightarrow 0, \end{array} \quad (3.2)$$

keeping the chain map a homotopy equivalence of exact sequences where the middle two maps are no longer isomorphisms, but the outer ones still are.

Repeating the procedure using Schanuel's lemma one final time with the short exact sequences

$$0 \rightarrow \ker g_2 \rightarrow E_2 \oplus D \xrightarrow{g_2 \oplus \operatorname{id}} \operatorname{im} g_2 \oplus D \rightarrow 0$$

$$0 \rightarrow \ker f_2 \rightarrow F_2 \oplus D' \xrightarrow{f_2 \oplus \operatorname{id}} \operatorname{im} f_2 \oplus D' \rightarrow 0$$

together with ψ gives another chain isomorphism

$$\begin{array}{ccccccc} 0 & \longrightarrow & \ker g_2 \oplus E' & \longrightarrow & (E_2 \oplus D) \oplus E' & \xrightarrow{(g_2 \oplus \operatorname{id}, 0)} & \operatorname{im} g_2 \oplus D \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow \wr & & \downarrow \psi \\ 0 & \longrightarrow & \ker f_2 \oplus F' & \longrightarrow & (F_2 \oplus D') \oplus F' & \xrightarrow{(f_2 \oplus \operatorname{id}, 0)} & \operatorname{im} f_2 \oplus D' \longrightarrow 0, \end{array}$$

where $E' = F_2 \oplus D'$ and $F' = E_2 \oplus D$, and gluing this to (3.2) results in a homotopy equivalence

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & \ker g_2 \oplus E' & \longrightarrow & (E_2 \oplus D) \oplus E' & \xrightarrow{(g_2 \oplus \text{id}, 0)} & E_1 \oplus D & \xrightarrow{(g_1, 0)} & E_0 & \xrightarrow{q_1} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \wr & & \downarrow \wr & & \downarrow & & \downarrow & & \downarrow \text{id} & & \\ 0 & \longrightarrow & \ker f_2 \oplus F' & \longrightarrow & (F_2 \oplus D') \oplus F' & \xrightarrow{(f_2 \oplus \text{id}, 0)} & F_1 \oplus D' & \xrightarrow{(f_1, 0)} & F_0 & \xrightarrow{q_2} & \mathbb{Z} & \longrightarrow & 0. \end{array}$$

We can then again cancel elementary complexes $(D)_*$ and $(D')_*$ from the rows to get a homotopy equivalence

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & \ker g_2 \oplus E' & \longrightarrow & E_2 \oplus E' & \xrightarrow{(g_2, 0)} & E_1 & \xrightarrow{g_1} & E_0 & \xrightarrow{q_1} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \wr & & \downarrow & & \downarrow & & \downarrow & & \downarrow \text{id} & & \\ 0 & \longrightarrow & \ker f_2 \oplus F' & \longrightarrow & F_2 \oplus F' & \xrightarrow{(f_2, 0)} & F_1 & \xrightarrow{f_1} & F_0 & \xrightarrow{q_2} & \mathbb{Z} & \longrightarrow & 0. \end{array}$$

But now this is precisely what we want, i.e. a chain map

$$\begin{array}{ccccc} E_2 \oplus E' & \xrightarrow{(g_2, 0)} & E_1 & \xrightarrow{g_1} & E_0 \\ \downarrow & & \downarrow & & \downarrow \\ F_2 \oplus F' & \xrightarrow{(f_2, 0)} & F_1 & \xrightarrow{f_1} & F_0 \end{array}$$

inducing isomorphisms on homology. Since all the modules are finitely generated and free (hence projective), this chain map is a homotopy equivalence by Theorem 1.59, as required. $\blacksquare$

Now we return to the inductive argument using F_* to construct a finite Y equivalent to X . The theorem is a special case of Theorem 2 from Wall [Wal66], and the proof we give follows Wall's, with added detail and altered base case (to utilise Lemma 3.6):

Theorem 3.8 ([Wal66, Theorem 2]). A CW complex X is homotopy finite if and only if $\pi_1(X)$ is finitely presented and $C_*(\tilde{X})$ is chain homotopy equivalent to a finite free chain complex.

Proof. The forward direction is trivial (by Proposition 1.4 and Corollary 1.60). For the other, suppose $f : F_* \rightarrow C_*(\tilde{X})$ is a chain homotopy equivalence with F_* a finite free complex. Apply Lemma 3.6 to the 2-skeleton of F_* to obtain Y 2-dimensional with $C_*(\tilde{Y}) \simeq (F_2 \oplus F' \rightarrow F_1 \rightarrow F_0)$ for some finitely generated free module F' . Then $F_* \oplus (F')^3$ is still a finite free complex, and it is still homotopy equivalent to $C_*(\tilde{X})$, so we can assume F_* to be of this form to begin with and then we have $C_*(\tilde{Y}) \simeq F_*^2$. Moreover, we know from the proof of Lemma 3.6 that Y is constructed from a presentation complex for $\pi_1(X)$ by attaching 2-cells to the basepoint: this allows us to define a map $g : Y \rightarrow X$ inducing an isomorphism on fundamental groups, by mapping all the generators of $\pi_1(Y)$ to generators of $\pi_1(X)$, whence all the relations in $\pi_1(Y)$ determine nullhomotopic loops in X , so the map can be extended over these; the additional 2-cells map to the basepoint. Since $C_*(\tilde{Y}) \simeq F_*^2$, we have that $H_2(\tilde{X}) \cong H_2(F_*)$ is a quotient of $H_2(\tilde{Y})$, and the portion of the homology exact sequence of $\tilde{g}$

$$H_2(\tilde{Y}) \rightarrow H_2(\tilde{X}) \rightarrow H_2(\tilde{g}) \rightarrow 0$$

has the first map surjective, hence $H_2(\tilde{g})$ is trivial and thus (by Hurewicz) so is $\pi_2(g)$, therefore g is 2-connected. We can now check that a homotopy equivalence between $\hat{g} : C_*(\tilde{Y}) \rightarrow F_*^2$ is actually induced by g (by composing $\tilde{g}_*$ with f^{-1} restricted to the 2-skeleton). This forms the base case for the induction.

Now suppose we have Y finite i -dimensional with $g : Y \rightarrow X$ i -connected and a homotopy equivalence $C_*(\tilde{Y}) \cong F_*^i$ induced by g . As in the proof of Theorem 2.10 we need $\pi_{i+1}(g) \cong H_{i+1}(\tilde{g})$ to be finitely generated, but this is not enough: we also need the new map $g' : Y' \rightarrow X$ to induce a homotopy equivalence $\hat{g} : C_*(\tilde{Y}') \rightarrow F_*^{i+1}$. So the cells we attach need to be induced by F_* using f .

Recall from Theorem 1.61 that $H_{i+1}(\tilde{g})$ is isomorphic to the $(i+1)$ -st homology module of the algebraic mapping cone $M_{*}^{\tilde{g}_*}$ of $\tilde{g}_* : C_*(\tilde{Y}) \rightarrow C_*(\tilde{X})$. $M_{*}^{\tilde{g}_*}$ is chain homotopy equivalent to the algebraic mapping cone of $f|_{F_*^i}$, i.e. we have

$$\cdots \rightarrow C_{i+2}(\tilde{X}) \rightarrow C_{i+1}(\tilde{X}) \oplus F_i \rightarrow C_i(\tilde{X}) \oplus F_{i-1} \rightarrow \cdots$$

and $H_{i+1}(\tilde{g})$ is the homology at the middle level. Compare this to $H_{i+1}(f) \cong H_{i+1}(M_*^f)$. This second is a quotient of $H_{i+1}(\tilde{g})$ by the image of F_{i+1} , and it is also trivial because f is a homotopy equivalence (here using Theorem 1.59). Thus the image of F_{i+1} is in fact the whole of $H_{i+1}(\tilde{g})$, and we have the required finite generation of $\pi_{i+1}(g)$ induced by f . Attach cells using Lemma 2.7 as usual, giving Y' finite and $(i+1)$ -dimensional with $g' : Y' \rightarrow X$ $(i+1)$ -connected, and the attachments of the cells being induced by f means that the isomorphism $C_{i+1}(\tilde{Y}') \cong F_{i+1}$ completes the chain map $\hat{g} : C_*(\tilde{Y}') \rightarrow F_*^{i+1}$, and this chain map induces an isomorphism on all homology, hence is a chain homotopy equivalence by Theorem 1.59 again.

As F_* is finite-dimensional, after finitely many steps all homotopy modules of g will be trivial, and we are done by Whitehead's theorem. $\blacksquare$

Now we are finally ready to round out the proof of the main result:

Theorem 3.1 ([Wal65, Theorem F]). A CW complex X is homotopy finite if and only if it is finitely dominated and $\omega(X) = 0$.

Proof. If X is homotopy finite, it is clearly finitely dominated and $C_*(\tilde{X})$ is a chain equivalent to a finite free complex F_* , hence $\omega(X) = \tilde{\chi}(F_*) = 0$.

On the other hand, if X is finitely dominated and $\omega(X) = 0$, then by Corollary 3.4 $C_*(\tilde{X})$ is chain homotopy equivalent to a finite free complex. The fundamental group of X is finitely presented, so X is homotopy finite by Theorem 3.8. $\blacksquare$

Chapter 4

The D2 problem

Now that we have proved our main theorem, we can take a step back and examine what the proof gives us. It is natural to ask, for example, for a given finitely dominated X , what the resulting finite complex Y equivalent to X actually looks like. In particular, if the dominating complex for X has dimension n , then what is the dimension of Y ?

We touched upon this at the end of Section 3.1, mentioning that Wall's original proof gives the dimension of the resulting Y . Now we will see how we can also obtain this information from our proof. The following proposition's proof compiles information from proofs in previous chapters and contains no new arguments:

Proposition 4.1. Let X be a CW complex dominated by a finite n -dimensional CW complex. If $\omega(X) = 0$, then X is homotopy equivalent to a finite CW complex of dimension $\max(3, n)$.

Proof. Let's carefully step through the construction of Y again, paying attention to dimension this time. First, in Theorem 2.10 and Corollary 2.11 we obtain a free chain complex of finite type Q_* with $Q_* \simeq C_*(\tilde{X})$; the dimension of the dominating complex plays no role here. Then in Theorem 2.1 we replace Q_* with a finite projective complex P_* with $P_* \simeq C_*(\tilde{X})$, and the dimension of P_* is n . Now since $\tilde{\chi}(P_*) = 0$, we use Theorem 3.3 to replace it again, this time with a finite free complex F_* , and the dimension of F_* is again n , except in the case $n = 0$ in which case we have that F_* is 1-dimensional. So far so good; we haven't strayed from the dimension of the dominating complex in the majority of cases.

We come to the proof of Theorem 3.8. Here, in the base case, in order to find a 2-dimensional Y realising the 2-skeleton of F_* , we first replace F_* with $F_* \oplus (F')_*^3$ for some finitely generated module F' : this has dimension $\max(3, n)$ in all cases. Then we attach cells to the 2-dimensional Y in accordance with the terms of F_* , so the final complex Y' indeed has dimension $\max(3, n)$. ■

This leaves us questioning whether we can actually achieve n -dimensional finite complexes in the cases $n < 3$. The case $n = 0$ is not interesting: all connected 0-dimensional CW complexes are homeomorphic, and counting connected components solves the problem in the disconnected case. So we have the following two problems:

1. If X is a CW complex dominated by a finite 1-dimensional complex and $\omega(X) = 0$, is X homotopy equivalent to a finite 1-dimensional complex?

2. If X is a CW complex dominated by a finite 2-dimensional complex and $\omega(X) = 0$, is X homotopy equivalent to a finite 2-dimensional complex?

Results of Stallings [Sta68] and Swan [Swa69] imply that a CW complex X dominated by a finite 1-dimensional complex with $\omega(X) = 0$ must have a finitely generated free fundamental group, whence by Wall [Wal65, Theorem D] X must be homotopy equivalent to a finite wedge of circles. This answers the first problem positively.

The second problem is a major open problem, known as the D2 problem:

D2 Problem. If X is a CW complex dominated by a 2-dimensional complex and $\omega(X) = 0$, is X homotopy equivalent to a finite 2-dimensional complex?

A lot of attention has been granted to the D2 problem, with multiple books devoted to the subject [Joh12, Joh03a], and also some recent progress [Ham19, Nic21]. Johnson [Joh03b] and Mannan [Man09] have shown the D2 problem to be equivalent to the following realisation problem:

Realisation Problem. Let G be a finitely presented group and $F_* = (F_2 \xrightarrow{f_2} F_1 \xrightarrow{f_1} F_0)$ be a finite free complex of $\mathbb{Z}[G]$ -modules with $H_1(F_*) = 0$ and $H_0(F_*) \cong \mathbb{Z}$ with the trivial $\mathbb{Z}[G]$ -action. Then does there exist a finite 2-dimensional CW complex Y realising F_* ?

This statement is precisely the statement of Lemma 3.6 but without the leniency toward adding on a finitely generated free module in dimension 2. In fact, that the D2 problem reduces to the realisation problem is almost evident from the work we have done: for, suppose the realisation problem had a positive answer; then we could use it in place of Lemma 3.6 in the proof of Theorem 3.8 and the dimension counting in Proposition 4.1 would work out to n for $n \geq 1$. The only hiccup is that the proof of Theorem 3.8 also relied on Y being constructed from a presentation complex for G to be able to define a 2-connected map $g : Y \rightarrow X$ inducing the homotopy equivalence $C_*(\tilde{Y}) \simeq F_*^2$; for details on how this is resolved, see Johnson [Joh03b].

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